Distinct critical network regimes facilitate memory encoding and consolidation

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Abstract: Systems operating near their critical point, or close to a transition between order and disorder, have computational advantages. In the case of neural networks, proximity to criticality may support optimal cognitive function. However, different cognitive processes rely on disparate computational demands. Using large-scale electrophysiological recordings in behaving rodents, we examined how critical dynamics in the hippocampus are regulated during learning and sleep-dependent memory consolidation. We found that operating near criticality enables learning by facilitating hippocampal coordination with input regions and maximizing flexibility of neural representations. In contrast, the hippocampal network shifts toward a more ordered, subcritical regime during sleep memory replay. In addition, proximity to criticality is restored by cholecystokinin interneurons-mediated inhibition. Our results provide a biological substrate of how critical dynamics in neural networks can support a variety of brain functions.

Systems closer to their critical point, that is, near a transition between order and disorder, are endowed with several advantages, including a higher dynamic range [1–3], enhanced computational capacity [4–7], and greater flexibility for information processing [8–11]. Thus, an influential proposal suggests that the brain operates close to criticality to achieve optimal performance [6,7,9,12–19]. Physiological signatures of critical dynamics, such as power law distributions, have been observed through *in vitro* preparations [2,8,20] and in coarse-grained neural recordings in animals [21–28] and humans [29–36]. Furthermore, several brain disorders are associated with a disruption in critical dynamics [14,17,18,37–41]. However, it is not yet known what the biophysical nature of criticality in the brain is. Furthermore, different cognitive processes require distinct computational demands. For learning new information, neuronal networks may benefit from higher flexibility or an ample dynamic range, both facilitated near the critical point. In contrast, consolidating recent experiences may require minimizing susceptibility to external perturbations, conditions which will be favored far from the critical point. In this study, we used large-scale electrophysiological recordings in behaving rodents to investigate the neural mechanisms that regulate critical dynamics in the CA1 region of the hippocampus during learning and sleep-dependent memory consolidation. We hypothesized that the hippocampal network adjusts dynamically across distinct critical regimes to support various cognitive processes. Because the critical point is thought to provide systems with a higher dynamic range [1–3], optimized computational capacity [4–7], and greater flexibility for information processing [8–11], we reason that the hippocampus would operate near criticality to optimize learning. Conversely, transitioning into a more subcritical regime during memory consolidation would favor a more ordered state that could facilitate memory replay (Fig.1a).

Hippocampal activity distance to criticality varies across brain states

Because memory encoding and consolidation happen during distinct brain states [42], we first examined hippocampal CA1 activity across waking and sleep (Fig.1). To quantify criticality, we used a recently developed method based on the temporal renormalization group (tRG) theory [43] (Fig.1b). Traditional approaches based on the detection of avalanches (population bursts surpassing a threshold) such as distance to criticality coefficient (DCC) or branching ratio (BR) (Extended Data Fig.1a) are computed over long periods of time, compromising the discovery of faster dynamic changes in criticality. An advantage of the tRG-based method is the use of autoregressive (AR) models to fit experimental data, which enables a time-resolved estimate of proximity to criticality (d_2).

As expected, the explained variance of the AR model fit was higher for shorter time windows and plateaued as window length increased (Extended Data Fig.1b), yielding comparable signal-to-noise (SNR) trends for 30- to 60-second windows consistent with previous reports [44] (Extended Data Fig.1c). To avoid introducing artifactual fluctuations in d_2, we excluded periods of state transitions, which involve abrupt shifts in firing dynamics (Extended Data Fig.1d).

We found that the hippocampus' distance to criticality fluctuated dynamically during awake and sleep states (Fig.1c), consistent with recent observations in cortical areas [25,28,43,45,46]. On average, the hippocampus operated closer to criticality during waking than during sleep (Extended Data Fig.1e). Specifically, the hippocampus was closer to criticality during active theta (Fig. 1d, left) and during REM (Fig. 1d, right) compared to wake non-theta and NREM sleep respectively.

Comparable results were obtained when analyzing the BR, which quantifies how much activity propagates from one time bin to the next (Extended Data Fig.1f).

Overall, during waking, CA1 firing rates were positively correlated with d_2 during theta and negatively correlated during non-theta periods (Fig.1e, left). In contrast, both REM and NREM sleep showed positive correlations between firing rates and d_2 (Fig.1e, right). Because one of the hallmarks of criticality is a higher dynamic range of the system [1–3], we next quantified the dynamic range of CA1 firing rates [2] in relation to d_2. Notably, despite state-dependent differences in d_2, we found a higher dynamic range of CA1 firing rates during both awake and sleep states (Extended Data Fig.1g, left and right respectively) when it was closer to criticality. Overall, these results indicate that hippocampus' critical dynamics fluctuate rapidly within each brain state and strongly differ across states, while maintaining a consistently higher dynamic range in proximity to criticality.

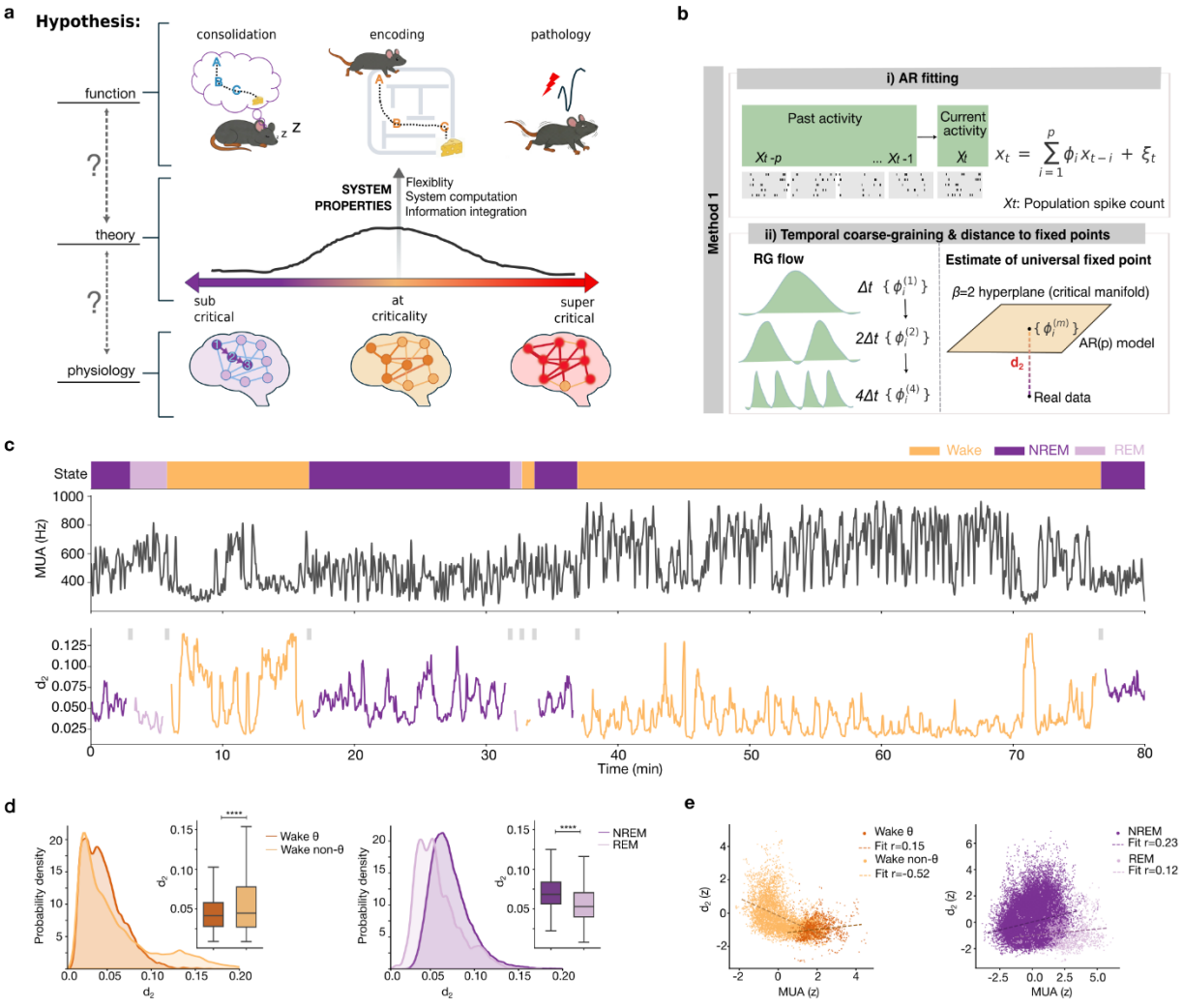


Fig. 1 a, Hypothesis: critical dynamics in the hippocampus fluctuate between a state close to criticality during memory encoding to a far from criticality, subcritical regime during memory consolidation. **b,** Overview of the methodological approach used to quantify distance to criticality. Top: autoregressive models, AR(p), are fit to the z-scored multiunit spike counts, using the temporal renormalization group (tRG) theory to extract the AR coefficients that are later used to provide a time-resolved estimate of the distance to criticality as d_2 (bottom), that is, the Euclidean distance from the fitted AR model to the $\beta = 2$ fixed-point hyperplane (considered the scale-invariant critical manifold). Note that lower and higher d_2 values imply closer and farther away from criticality respectively. Grey rectangles denote periods around state transitions during which d_2 was not computed. **c,** Example of continuous recording illustrating state segmentation (Wake, NREM, REM; top), multiunit firing rate (middle), and d_2 to criticality (bottom). **d,** Left: distribution of d_2 during waking for active (theta, θ) and non-active (non-theta, non-θ) periods. Inset boxplots show group comparisons of d_2 value ($p < 1\times10^{-15}$, paired linear mixed-effect model (LMM), n = 18 sessions from n = 6 animals). Right: same for sleeping periods of NREM and REM states. Inset boxplots show group comparisons ($p < 1\times10^{-15}$, paired LMM, n = 18 sessions from n = 6 animals). **e,** Relationship between d_2 and multiunit activity (MUA) during theta and non-theta states during wake (left panel) and REM and NREM stages during sleep (right panel). The d_2 relationship with MUA displayed a positive ($r = 0.15$, $p < 1\times10^{-15}$) and a negative ($r = -0.52$, $p < 1\times10^{-15}$) correlation during theta and non-theta wake states respectively, whereas both NREM and REM sleep stages showed a positive relationship (NREM: $r = 0.23$, $p < 1\times10^{-15}$; REM: $r = 0.12$, $p = 1.17 \times 10^{-5}$). Pearson correlations were used for all comparisons, n = 18 sessions from n= 6 animals). Asterisks denote significance (*p<0.05, **p<0.01, ***p<0.001, ****p<0.0001; n.s., not significant).

The hippocampus moves away from criticality during replay of recent experiences

Because the hippocampus moved away from criticality during sleep, we sought to understand whether this departure was related to sleep-dependent memory consolidation processes, as we originally hypothesized (Fig. 1a). To answer this, we first compared d_2 values in sessions of pre- versus post-task sleep in mice that trained on a hippocampal-dependent spatial memory task [47]. We found that d_2 distance increased during post- compared to pre-task NREM sleep (Fig. 2a). This pre- to post-task NREM sleep increase was more pronounced after animals trained in a novel maze than following training on a familiar maze (Extended Data Fig.2a), suggesting that higher memory load displaced the network further from criticality.

A well-established mechanism for sleep-dependent memory consolidation is the replay of neural sequences corresponding to activity patterns expressed during recent behavior [47–53]. Such memory replay is orchestrated by hippocampal sharp-wave ripples (SWRs) [49,54], whose incidence increases after learning (Extended Data Fig.2b) [53,55]. Therefore, we detected replay events (Fig. 2b) during sleep using a Bayesian decoder trained on the CA1 activity patterns recorded during the behavioral task [47,56]. Approximately 22 % of SWRs were associated with significant replay (Fig. 2b), consistent with previous reports [47,52,56–59].

Interestingly, we observed that replay probability co-fluctuated with the hippocampus' distance to criticality during NREM sleep (Fig. 2c), and that the two measures were significantly correlated (Fig. 2d). Conversely, d_2 was positively correlated with the rate of replay events but not with the

rate of SWRs lacking significant replay (Fig. 2e), and this dissociation couldn't be explained by the associated d_2 uncertainty [44] within SWR's sextiles (Extended Data Fig.2c).

To quantify the relative contribution of different variables to replay probability, we built a generalized linear model (GLM) (Fig. 2F, top). The full model *(whole*) was fitted with a combination of distance to criticality (d_2), mean multiunit firing rate (*FRm*) and multiunit firing rate variability across neurons (*FRstd*) during NREM sleep. Ablation models were generated by removing different variables. The reduction in predictive gain was significantly greater when d_2 was ablated compared to *FRm* or *FRstd* ablation (Fig. 2F, bottom), suggesting that proximity to criticality is a strong predictor of replay probability.

As an alternative approach of assessing criticality, we quantified avalanche size and duration (Extended Data Fig.2d), power-law scaling properties (Extended Data Fig.2e) and DCC (Extended Data Fig.2f) [25,45]. For all three measures, we found that NREM periods of high-SWR rate, during which replays were most prominent [47], were farther from criticality than periods of low-SWR rate (Extended Data Fig.2d, e, f).

Next, we asked whether the network shifted towards sub- or super-critical regimes around SWRs. To answer that, we calculated the BR (Extended Data Fig.1a), which quantifies how activity propagates from one time bin to the next and classifies dynamics as subcritical (BR < 1), critical (BR ≈ 1), or supercritical (BR > 1) [8,24,60]. The BR in CA1 showed subcritical values and significantly decreased during periods of high SWR rate compared to shuffles (Fig. 2G). Because subcritical regimes are characterized by lower entropy compared to critical regimes [9,11], we next quantified the entropy of CA1 firing dynamics across different SWR rates. We found that entropy was significantly lower during periods of higher SWR rate (Fig. 2h).

Our observation that the hippocampus moves away from criticality during post-task NREM sleep raises two possible interpretations. One is that the structured population spiking bursts associated with memory replay drives the observed elevated d_2 values. Alternatively, the observed increase of d_2 values could reflect a global transition into a subcritical regime, which may itself be a prerequisite for replay. To distinguish between these possibilities, we first checked for the increased SWR rate after learning (Extended Data Fig.2b) by comparing pre- and post-task NREM periods of matched SWR rate distributions. We found that d_2 remained higher in post- compared to pre-task NREM sleep after controlling for SWR rate (Extended Data Fig.3a). This result persisted even after excluding all spikes within SWRs and analyzing only the remaining activity within the time window (Extended Data Fig.3b). Similarly, the positive correlation between SWR rate and d_2 remained after removing all spikes within SWRs (Extended Data Fig.3c). Similarly, the entropy of the firing dynamics continued to show a negative correlation with SWR rate after removing SWR firing (Extended Data Fig.3d). Altogether these results show that during post-task NREM sleep, CA1 transitions into a highly subcritical regime that provides a permissive network state for the emergence of memory replay during SWRs in support of memory consolidation.

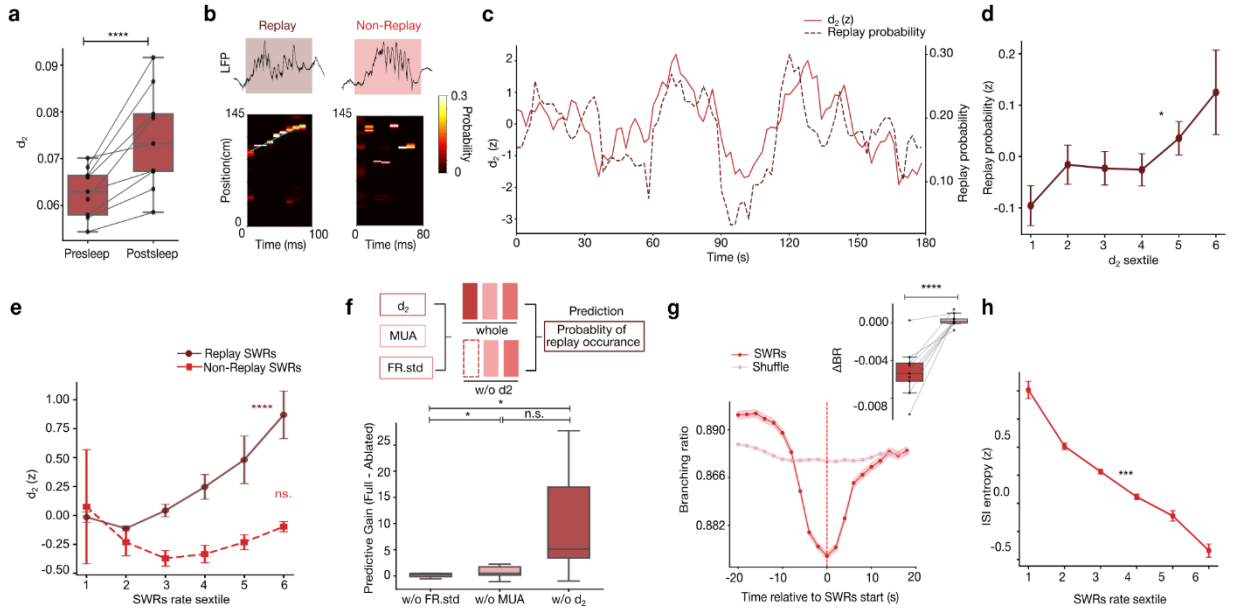


Fig. 2 The hippocampus moves away from criticality during memory replay. a, d_2 increases from pre- to post-task sleep ($p = 2.36 \times 10^{-6}$, n = 9 sessions from n = 6 animals, paired LMM). **b,** Example of SWRs associated with replay (ordered firing, left) and non-replay (no preferred order, right). For each event CA1 LFP and decoded position (the cyan line denotes the best linear fit) for the event are shown **c,** Example trace of z-scored d2 (red, left Y axis) and replay probability (dashed dark red, right Y axis), computed in 30-s sliding windows. **d,** Z-scored replay probability (mean ± SEM) per d_2 sextiles of NREM sleep. Note that replay probability increases with higher d_2 values ($p = 0.016$, n = 9 sessions from n = 6 animals; linear mixed-effect model). **e,** d_2 correlated with the rate of SWRs associated with replay but not with the rate of SWRs not associated with memory replay ($p < 10^{-15}$ and $p = 0.80$ for replay and non-replay curves respectively, n = 9 sessions from n = 6 animals, linear mixed model). **f,** Top: schematic of the logistic regression framework used to predict replay occurrence from predictors (d_2, MUA, FR_std) while allowing variability of feature effects across bins as a feature×state interaction term. Bottom: predictive gain for each feature interaction term, quantified by the change in cross-validated performance upon feature ablation. Post hoc paired Wilcoxon tests (Holm-corrected) indicated d_2 > MUA ($p = 0.035$) and d_2 > FR_std ($p = 0.039$), while MUA ≈ FR_std ($p = 0.91$). **g**, Branching ratio (BR) during SWRs and shuffled periods for an example session. Ripple windows showed a significant decrease in BR compared to shuffles ($p < 1\times10^{-15}$, n = 9 sessions from n = 6 animals, paired LMM). **h,** Inter-spike-interval (ISI) entropy decreased with ripple rate (Pearson's: r = -0.99, $p = 3.57\times10^{-4}$, n = 9 sessions from n = 6 animals).

BARRs contribute to restoring the hippocampal network to its critical point

Because we observed that the hippocampus moved away from criticality during memory consolidation (Fig.2), we next asked how the hippocampal network restores its proximity to criticality. To identify potential mechanisms able to restore proximity to criticality, we built a simplified CA1 network model. To investigate how the activity of distinct neuronal populations influences critical dynamics beyond excitation and inhibition, the model included pyramidal cells (E), fast-spiking parvalbumin (PV) interneurons, and regular-spiking cholecystokinin-expressing

(CCK) interneurons [61–65]. In addition, it featured excitatory inputs from CA3, known to be important for SWRs generation, targeting all cell types in CA1 [64] and from CA2 targeting CA1 CCK interneurons [53] (Fig. 3a).

The model reproduced the relationship between multiunit firing rate and d_2 experimentally observed during sleep (Extended Data Fig.4, Fig. 3b and Fig.1 e, right). The strength of CA3 input was positively correlated with d_2 in CA1 (Fig. 3c), consistent with our experimental results during SWRs (Fig. 2d, e and Extended Data Fig.3f). In contrast, the strength of CA2 input onto CCK interneurons was negatively correlated with d_2 (Fig. 3d). This result led us to hypothesize that CCK-mediated inhibition in CA1 may provide a mechanism for restoring the network's proximity to criticality.

Recently, we showed that a circuit from CA2 pyramidal cells to CCK basket cells in CA1 generates barrages of action potentials (BARRs) during sleep to counterbalance the elevated firing rates associated with memory replay (Extended Data Fig.4a) [53]. Therefore, we next asked whether BARRs contribute to restoring the network's proximity to criticality. To answer that, we performed simultaneous recordings from CA1 and CA2 regions during the same behavioral paradigm and following the same task structure than before (Fig. 3e-h). We identified BARR events during post-task sleep, which, as expected, were accompanied by increased CA2 pyramidal cell activity (Extended Data Fig.3g). BARR rate was inversely correlated with d_2 during post-task NREM sleep (Fig. 3E), in line with the predictions from our model (Extended Data Fig.4b-d, e-g, h-j). Similarly, the BR significantly increased during BARRs compared to shuffles (Fig. 3f), even when removing all spikes within BARRs (Extended Data Fig.3e), and the entropy of the CA1 firing dynamics was larger during periods of higher BARRs' rate (Fig. 3h).

To causally test the contribution of CCK-mediated inhibition to network dynamics, we next looked at sessions during which a subset of CA1 CCK interneurons was optogenetically silenced during BARRs in post-task sleep [53]. In these sessions, the CA1 network remained farther away from its critical point compared to control sessions (Fig. 3g). Together, these results demonstrate that CCK-mediated inhibition during BARR states contribute to restoring the hippocampal network's proximity to criticality.

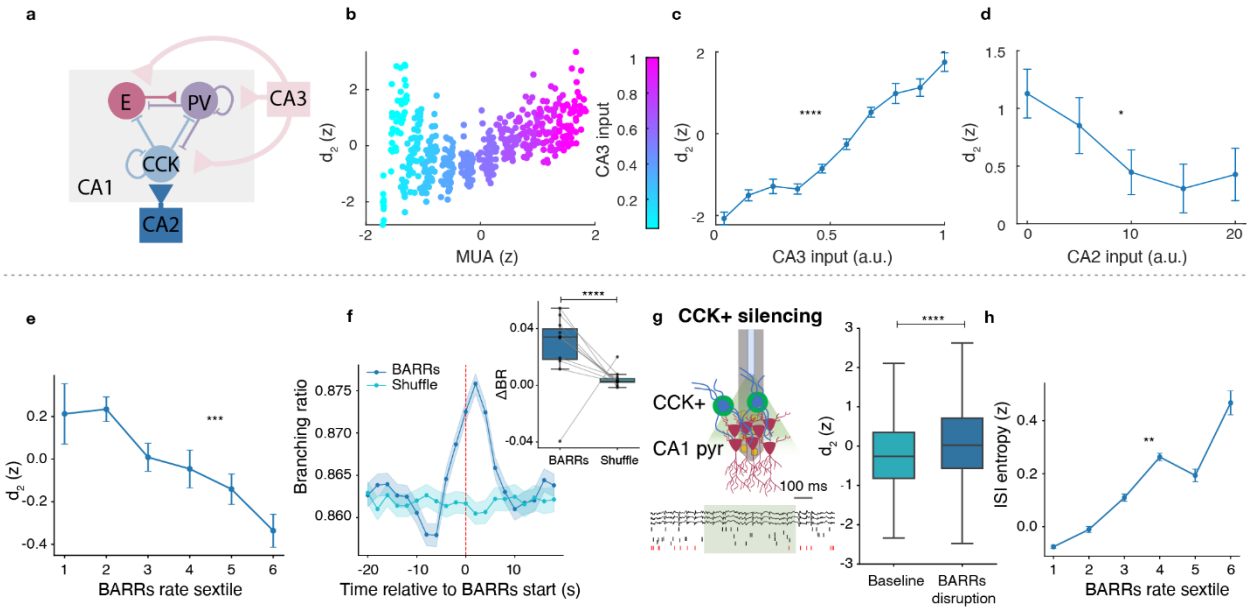


Fig. 3 CCK inhibition during BARRs contributes to restoring criticality in the hippocampus. **a,** Simplified network model of CA1, featuring excitatory (lines with triangles) and inhibitory connections (T-lines), and neuronal subpopulations in different colors. The model comprised one recurrent layer with a pyramidal neurons' population (E) and two interneuron populations (PV and CCK), as well as two feedforward inputs from CA3 and CA2, modeled as independent Poisson units, providing excitatory input to the CA1 network. **b,** In the model, CA1 d_2 was significantly correlated with multiunit firing rate (Spearman 's: $\rho = 0.92$, $p < 10^{-15}$), which was in turn modulated by the strength of CA3 input. **c,** CA1 d_2 increased with the strength of CA3 input ($p < 10^{-15}$, $r = 0.92$, Spearman correlation). **d,** CA1 d_2 decreased with the strength of CA2 input ($p = 0.011$, $r = -0.35$, CA3 Input = 0.9, Spearman correlation). **e,** In mice, CA1 d_2 was negatively correlated with the rate of barrages (BARRs) (Pearson's r = -0.969, p = 0.0014 mean ± SEM). **f,** Branching ratio (BR) during BARRs and shuffled periods for an example session. BARR windows showed a significant increase in BR compared to shuffles ($p = 2.71 \times 10^{-7}$, n=11 sessions from n = 5 animals, paired LMM). **g,** Left: schematic of optogenetic inhibition of CA1 CCK basket cells during sleep. Right: d2 stayed larger in sessions with CCK inhibition compared to control sessions ($p < 10^{-15}$, n=7 session from n = 3 animals, paired linear mixed model). **h,** Inter-spike-interval (ISI) entropy was positively correlated with the rate of BARRs (Pearson's: r = 0.946, p = 0.0043, n = 12 session from n = 5 animal, mean ± SEM).

Proximity to criticality of hippocampal activity facilitates learning

Next, we sought to test our prediction that proximity to criticality facilitates learning (Fig.1a). To do that, we turned to a spatial task that enabled us to assess de novo learning each day. We performed silicon probe recordings of all subregions of the hippocampus and medial entorhinal cortex (MEC) in rats during a spatial learning task in a cheeseboard maze. In this task, rats had to learn the location of three hidden water rewards whose configuration changed daily [59,66–68], followed by ~1.5 hours of sleep in their home-cage (Fig.4a). A delay period in the start-box was imposed between trials during learning. On the following day, animals underwent a recall test to evaluate memory for the previous day's reward locations [59,66–68] (Fig.4a).

We replicated our previous findings in this new dataset, including that CA1 operated closer to criticality during theta compared to non-theta waking periods, and during REM relative to NREM sleep (Extended Data Fig.5a, b). The dynamic range of CA1 firing rates was also higher when the hippocampus was closer to criticality (Extended Data Fig.5c, d). In addition, we observed that the hippocampus departed and returned closer to criticality around SWRs (Extended Data Fig.6a-d) and BARRs (Extended Data Fig.6e-g) respectively, paralleled by a decrease and increase of the neuronal inter-spike-interval entropy.

During the task, rats engaged in random search during the earlier trials and quickly learned to take directed trajectories towards rewarded locations (Fig. 4B). We hypothesized that a network regime closer to criticality would benefit subsequent learning on this task. To test this, we compared behavioral performance (quantified as path length to collect all rewards) with CA1 distance to criticality measured across different task phases.

We found that proximity to criticality during learning trials was positively correlated with performance (Fig.4c and Extended Data Fig.7). Overall, the hippocampus was closer to criticality during earlier compared to later trials (Fig.4d), and these effects could not be explained by speed, firing rate or theta power (Extended Data Fig.7). Moreover, the hippocampus' distance to criticality during inter-trial intervals in the start-box correlated with subsequent trial performance (Fig.4e), and inter-trial intervals during earlier trials were closer to criticality compared to later trials (Fig.4f).

To assess the relative contribution of different variables to learning performance, we constructed a generalized linear model (GLM) (Fig.4g, left). The full model *(whole*) incorporated multiunit firing rate (*FRm*), animal speed (*speed*) and distance to criticality (d_2) during the task. We then generated ablation models by removing each one of the predictors. We found that the largest drop in predictive gain between the whole and the ablated models was obtained for d_2 (Fig.4g, right), indicating that proximity to criticality strongly contributes to predicting behavioral performance.
Finally, we asked whether these results were specific to learning or instead reflected a general property of the waking state. To address this, we compared home-cage waking periods with inter-trial start-box intervals. The CA1 network was consistently closer to criticality during start-box periods than during home-cage waking epochs of the same duration (Fig.4h). In addition, proximity to criticality no longer predicted performance during next day recall trials (Fig.4i) after animals had already learned the reward configuration, in contrast to the learning phase (Fig. 4c, d).

To investigate the neural circuit mechanisms linking criticality in the hippocampus with behavioral performance, we examined how strongly CA1 coordinated with upstream inputs. To quantify cross-regional coordination between different areas, we computed canonical correlation analysis (CCA) between CA1 and CA2, CA3, or MEC regions (Extended Data Fig.8a) during the earlier learning trials (Fig. 5b). CCA between MEC and CA1 showed a negative correlation with distance to criticality, indicating a higher coordination between MEC and CA1 when the CA1 network was closer to criticality. A similar result was found between CA1 and CA3, as well as between CA1 and CA2 (Extended Data Fig.8b, c-e). Overall, these results show that CA1 network can more effectively coordinate with input regions when operating closer to criticality.

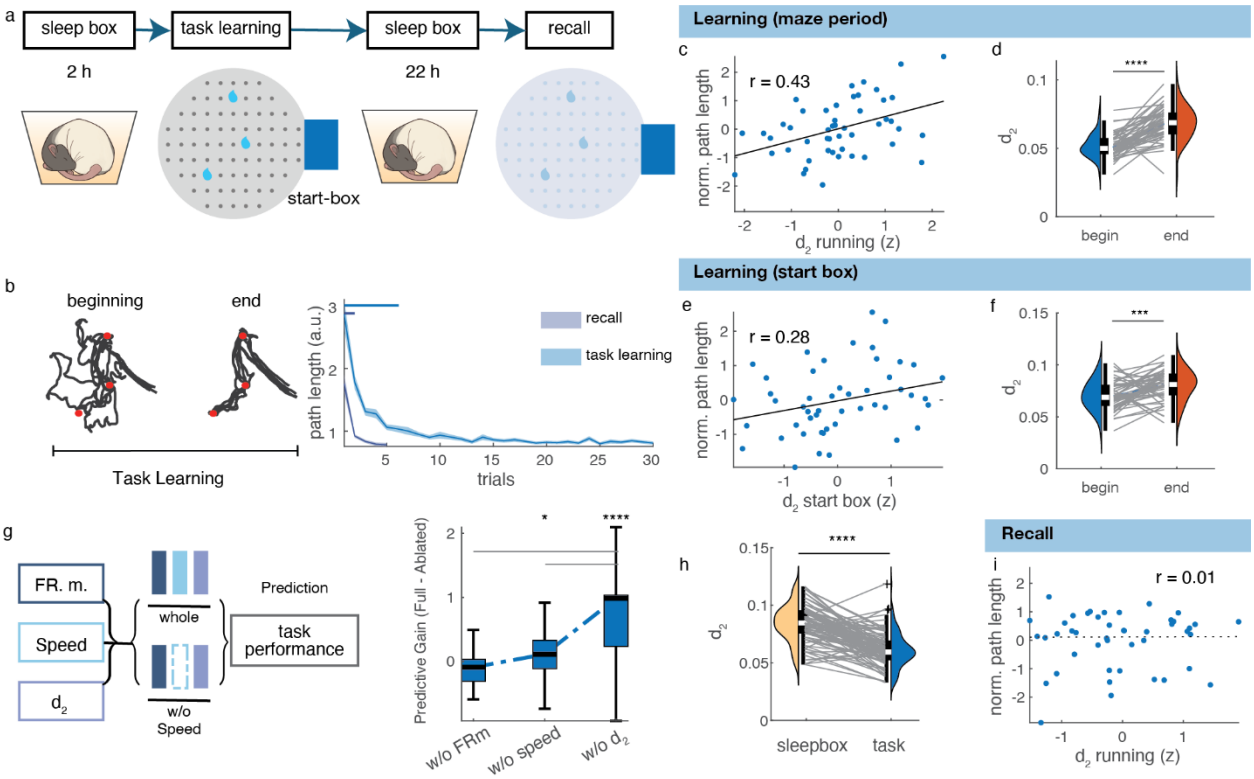


Fig. 4 Hippocampus proximity to criticality predicts behavioral performance. a, Schematic of the cheeseboard task. Rewards' location changed daily, forcing the animals to learn a new path each day, and memory consolidation was tested 22 hour later. **b,** Left: example trajectories in early and late trials. Red dots = reward locations. Behavioral performance (assessed as mean path length) during the task learning and recall phases. Animals learned the optimal trajectory to collect all three rewards within the first few trials of the task ($p < 10^{-15}$, n = 49 sessions from n = 3 animals, Kruskal-Wallis test with post-hoc Dunn test and Sidák's adjustment against last trial) and first 2 trials during the recall phase ($p < 10^{-15}$, n = 45, Kruskal-Wallis test with post-hoc Dunn test and Sidák's adjustment against last trial). **c,** Behavioral performance was correlated with d_2 during the earlier trials (first 5 trials) of the learning phase (r = 0.43; Spearman: r = 0.39, $p = 5.90\times10^{-3}$). **d,** d_2 was lower in the earlier (first 5 trials) compared to later (last 5 trials) trials during the learning phase of the task ($p=1.50\times10^{-8}$, n = 49, Wilcoxon signed rank test). **e,** Behavioral performance was correlated with d_2 during the preceding start-box periods (r = 0.28; Spearman: r = 0.32, p = 0.026). **f,** d_2 during start-box periods was significantly lower in the earlier compared to later trials ($p = 9.46\times10^{-4}$, n = 49, Wilcoxon signed rank test). **g,** Left: Schematic of the GLM model. Right: prediction gain of the full model compared with different ablation models. Note the strongest reduction of predictive gain for d2 (*p=0.023, ****$p= 1.316\times10^{-13}$, Wilcoxon signed rank test). **h,** d_2 during the task was lower compared to sleep-box waking periods of equal duration (5 minutes periods; $p < 0.0001$, Wilcoxon signed rank test, n = 90 sessions from n = 4 animals). **i,** Behavioral performance was not correlated with d_2 during the recall phase (r = 0.01; Spearman: r = 0.3, p = 0.84).

Distinct hippocampal representations emerge near criticality

Next, we sought to understand the computational mechanism through which proximity to criticality benefits learning. An additional hallmark of systems close to criticality is increased flexibility, that is, the capacity to access a broader range of trajectories in state space in response to inputs [4,11,69]. Recently, we showed that the hippocampus exhibits a similar form of flexibility, generating distinct latent representations of the same environment upon contextual changes, such as different reward configurations [68]. To investigate how criticality relates to such representational flexibility, we trained rats in a dual-configuration version of the original cheeseboard maze task [68]. In this task, rats were trained to collect three hidden rewards located in fixed locations over ~20-30 trials, followed by a sleep session in the home-cage. They were then introduced to a new reward configuration for another ~20-30 trials (Fig.5a). Sleep sessions preceded and followed each training session, and a recall trial was run 20 hours (next day) after the training in the second configuration.

To characterize the degree of hippocampal representational flexibility, we used dimensionality reduction techniques to extract neural representations of the two different reward configurations [68]. Specifically, we looked at CA1 population activity during learning trials using time contrastive learning implemented with CEBRA (Consistent Embedding for Bipartite Relationship learning with Auxiliary variables) (Fig.5b), a self-supervised machine learning method to reduce neural activity into a low-dimensional manifold [70]. Two different maps of the same maze corresponding to the two reward configurations emerged at the population level, consistent with recent findings [68,71] (Fig.5c), while single cell responses showed only moderate changes across maze configurations (Extended Data Fig.9a). These maps, or neural manifolds, displayed significantly higher consistency than shuffles (Fig.5d). Similar results were found when assessing manifolds' separability by training a support vector machine (SVM) to separate the two configurations versus shuffles (Fig. 5e), and the separability quickly saturated when considering low latent dimensions (Extended Data Fig.9c).

Better behavioral performance in the second session of the task was associated with greater separability of the manifolds of the two task configurations, both during learning trials (Fig.5f) and during inter-trial intervals (Fig. 5g), and these results couldn't be explained by a trivial multiunit firing rate relationship (Extended Data Fig.9d, e).

In addition, the separability of the different maps was larger when the hippocampus operated closer to criticality, whether assessed during learning trials (Fig. 5h) or during inter-trial intervals in the second maze session (Fig. 5i). Importantly, this relationship between proximity to criticality and separability was not present during the first maze session for either running trials or inter-trial intervals in the start-box (Extended Data Fig.9 h, iI). Similarly, proximity to criticality during a recall session preceding the learning during the earlier trials before the first maze did not predict map separability (Extended Data Fig.9 f, g). Together, these findings demonstrate that greater representational flexibility, quantified as differentiation of neural representations in CA1, emerges in proximity to criticality.

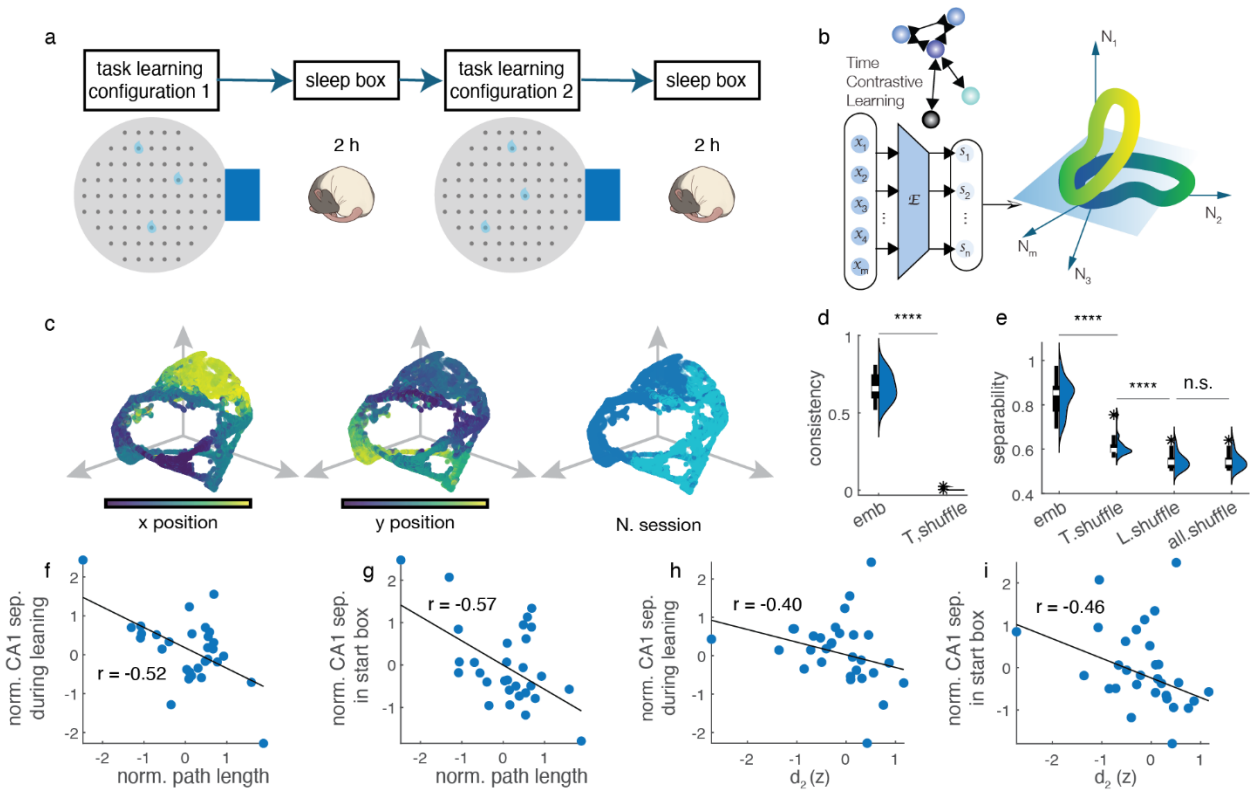


Fig. 5 Flexibility of hippocampal representations emerges near criticality. a, Schematic of the two-configurations cheeseboard task. **b,** A time contrastive learning-based method was used to find the latent hippocampal representations of the different task configurations. **c,** Left and middle: color coded manifold for X and Y positions in the cheeseboard maze. Right: Light and dark blue colors representing the two maze configurations. **d,** Latent embedding consistency is significantly higher than time shuffled embedding. latent manifold consistency measured as R^2 of the linear fit between 20 times of randomly trial shuffled embeddings ($p = 2.70\times10^{-5}$, n = 30, Wilcoxon signed-rank test). **e,** A support vector machine (SVM) was trained to separate manifolds representing the two configurations. Separability was measured as accuracy for each session with 5-fold cross-validation and compared to separability based on time sample shuffled embeddings, task label shuffled data, or time and label shuffled data ($p = 1.82\times10^{-5}$, n.s. : p = 0.63, Wilcoxon signed-rank test, n = 30 sessions). **f,** Separability of neural representations in CA1 during learning phase of the second configuration was correlated with behavioral performance ($r = -0.52$; Pearson: $r = -0.58$, $p = 8.54\times10^{-4}$, n = 30 sessions). **g,** Separability of neural representations in CA1 during start-box periods in the second configuration was correlated with behavioral performance ($r = -0.57$; Pearson: $r = -0.54$, $p = 2.18\times10^{-3}$, n = 30 sessions). **h,** Separability of CA1 maps during the learning phase of the second configuration was correlated with d_2 ($r = -0.33$, Spearman: $r = -0.40$, $p = 0.029$, n = 30 sessions). **I,** Separability of CA1 latent embeddings during start-box periods in the learning

phase of the second configuration was correlated with d_2 ($r = -0.46$, Spearman: $r = -0.40$, $p = 0.015$, $n = 30$ sessions).

Discussion

In this study, we investigated how critical dynamics in the hippocampal CA1 network contribute to the circuit mechanisms of memory encoding and consolidation. We hypothesized that a neuronal network close to criticality would facilitate memory encoding during behavior, while subcritical regimes would favor the occurrence of memory replay during sleep (Fig.1a). First, we found that criticality in the hippocampus fluctuates across and within brain states following a faster temporal dynamic than previously shown [25,45]. The hippocampus resided closer to criticality during awake compared to sleep (Fig.1), in line with recent results from cortical activity [43,72,73]. During sleep, we found that the hippocampus moved into its most subcritical regime during memory replay (Fig.2), suggesting that replay generation could be facilitated by a transient ordered state, which can be achieved in a subcritical domain. In support of this, removing all firing during SWRs but preserving the windows around them, did not change CA1 distance to criticality, suggesting that a subcritical state provides a permissive dynamic regime that facilitates replay generation. Using a biophysically realistic model of CA1, we predicted that the proximity to criticality could be restored by an increased activity of CCK interneurons (Fig.3, top). We found that during BARRs, hippocampal network states that entrain a circuit from CA2 to CA1 CCK interneurons [53], the hippocampal network relaxes closer to criticality (Fig. 3, bottom). This suggests that BARRs may serve as a homeostatic mechanism for restoring an optimal computational regime, better achieved close to criticality.

Animals' learning performance correlated with proximity to criticality, and this correlation disappeared once animals had already learned (Fig.4). The contribution of the critical regime to efficient learning could be driven by the enhanced dynamic range of firing rates (Extended Data Fig.1g), and better coordination with input areas, hallmarks of systems working close to criticality. In support of this, we found a higher coordination of CA1 activity with input regions, including MEC, CA3, and CA2, near criticality (Extended Data Fig.8). Another proposed advantage of systems that operate near criticality is higher flexibility or access to multiple system states [4,11,69]. Indeed, we found that representational flexibility, quantified as distinct neural maps associated with different contexts, emerged when the hippocampus operated near criticality (Fig .5). Such neural flexibility in the hippocampus can enable the integration of new memory representations into an existing repertoire [74], a mechanism that could facilitate continuous learning [5,75,76].

Importantly, although the hippocampus featured distinct critical regimes during memory encoding and consolidation, both states remained below the critical point. In contrast, the hippocampus entered a supercritical regime during epileptic seizures (Extended Data Fig.10), underscoring the importance of optimal critical dynamics.

In summary, our study establishes a link between the free-scaling properties of a physical system, assessed by critical dynamics, and the biological substrate, assessed by specific circuit mechanisms, that supports brain functions. We showed that memory encoding and consolidation alternate between the vicinity of the critical point and a subcritical regime to balance

computational demands and network homeostasis. Our findings suggest that optimal learning systems, whether biological or artificial, may require a dynamic regulation between flexible and rigid states, and can offer biophysical constraints to guide the design of Large Language Models (LLM) tuned to criticality, a current solution used to overcome the problem of information scarcity in artificial intelligence (AI) [77–79].

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Competing interests: Authors declare that they have no competing interests.

Data and materials availability: All data are available in the main text or the supplementary materials. Custom scripts used in this study can be downloaded from https://github.com/ayalab1/neurocode.

Supplementary Materials

Materials and Methods

Experimental model details

All experiments conformed to guidelines established by the National Institutes of Health and have been approved by Cornell University's Institutional Animal Care and Use Committee. Mice were kept in the vivarium on a 12-hour light / dark cycle with *ad libitum* access to food and water except during training. They were housed with a maximum of 5 per cage before surgery and individually afterward. C57BL/6J and B6FVBF1/J male mice (~27-32 g, 3-6 months old, The Jackson Laboratory) were used for electrophysiological recordings, and Sncg-Flp/+ [65] injected with Flp-dependent archaerhodopsin (AAV-nEF-Coff/Fon-Arch3.3-p2a-EYFP) mice (~26-30 g, 3-8 months old) were used for optogenetic manipulations. Rats (n = 7; adult male Long-Evans, 300-500 g, 3-6 months old) were kept in the vivarium on a 12-hour light/dark cycle. They were housed with a maximum of 2 per cage before surgery and individually afterward.
Temperature and humidity in the rooms were kept at 68-72 F and 40-60%, respectively for both mice and rats.

Surgical procedures

A variety of silicon probes were used in these these recordings, includingA5x12/16-buz/lin-5mm-100-200-160(64ch), ASSY-156-P1(64ch), A4x32-Poly2-5mm-23s-200-177(128ch). All silicon probes were manually glued to micro-drives to allow fine adjustment of their vertical location after surgery. For electrodes used for some of the optogenetic manipulations, an optic fiber 200µm (Thorlab FT200UMT) was manually glued to one of the shanks under the microscope. Electrodes were implanted in the hippocampus following previously described protocols [47,52,53,59,67,80–82]. In brief, animals were anesthetized with isoflurane anesthesia, the surgical area was shaved, skin was washed with layers of iodine and alcohol, and a clean single-cut incision was made in the skin to expose the skull and take coordinates. Coordinates to target the hippocampus were marked using stereotaxic micromanipulators. Next, a layer of Metabond was applied to the full extent of the skull except through the marked craniotomy's location, and a customized 3D-printed plastic base plate was glued to the skull using Metabond and dental acrylic. The base plate had four flaps of copper mesh previously attached that were left lying over the sides of the implant area to later cover the implant. Once dried, one stainless-steel wire or screw connected to a thin wire (California wires) was inserted over the cerebellum, cemented in place using C&B Metabond (Parkell) and used as a common ground and reference for the recordings. Following next, hippocampus craniotomies were performed, cleaned and dura removed. The probe(s) was then inserted above the dorsal hippocampus or medial entorhinal cortex (MEC) using stereotaxic micromanipulators (mice coordinates: AP = -1.94 mm, CA1: ML ~1.5-2.0 mm, CA2: ML: ~1.5-2.5, implantation depth = ~1 mm from the surface of the brain) ; rats coordinates: AP = -3.7 mm, CA1: ML ~2.5 mm, CA2: ML: centered at ~4.0, implantation depth = ~1.8 from the surface of the brain, MEC: AP = -6.0 mm, ML = 4.5 mm, implantation depth = ~3 mm from the surface of the brain). After the probes were lowered into their final implantation depth, the drive was cemented to the skull, micromanipulators retracted and

craniotomies sealed with artificial dura-gel (DOWSIL™ 3-4680 Silicone Gel Kit). Finally, the four flaps of the copper mesh were bent upwards forming a "hat" over the head of the animal and soldered between them through light aluminum bars to provide stability to the implant. The ground wire placed over the cerebellum then connected to the ground and reference wires of the electrodes and altogether to the copper mesh. Immediately after surgery animals were moved to their home-cage, which was placed with half of the area over a heating pad until animals were fully awake, moving and drinking/eating, after which the home-cage was moved to their corresponding holding room. Post-operative care including carprofen subcutaneously for 3 days (mice) or one week (rats) as well as triple antibiotic ointment over the wounded area starting the day of the surgery. After full recovery (~1 week), the probe was slowly lowered in small steps of 75–150-μm per day until the final position was targeted. Physiological landmarks based on local field potentials (LFPs) and unit firing patterns were used to identify the hippocampal pyramidal layers[47,52,53,59,67,80–82].

For BARR optogenetic manipulations, a previous procedure involving virus injection into the hippocampus was performed. Viral injection procedures were performed 2-3 weeks before electrodes were implanted. Briefly, mice were anesthetized with isoflurane anesthesia, the surgical area was shaved, skin was washed with layers of iodine and alcohol, and a clean single-cut incision was made in the skin to expose the skull and take coordinates. Coordinates to target the hippocampus were marked using stereotaxic micromanipulators and the craniotomy was performed. A glass micropipette had been previously loaded with mineral oil and then with the virus to be injected using an automatic injector (Nanoject III). The glass pipette was driven into the skull and using the surface of the brain as a reference, lowered until the desired location (coordinates: AP = -1.94 mm, ML ~1.5-2.0 mm, implantation depth = ~1 mm). Automatic injection of the virus was then started and performed at low speed, after which the pipette was left unmoved for ~15 mins to prevent interference with virus local diffusion. After the pipette was retracted, the skin was sutured and the wound treated with iodine and triple ointment antibiotic. Animals were injected with carprofen and placed in their home-cage with half of the area over a heating pad until full recovery, after which they were moved to their corresponding holding room.

Behavioral tasks

Mice behavioral tasks:

Before surgery, animals were accustomed to one maze for 3-5 days. After surgery recovery, mice were handled daily and accommodated to the experimenter, recording room, and cables for at least a week before the start of the experiments. One day before the beginning of the training, mice were placed under water restriction. All recording sessions included a session of sleep before and another session after the performance of the task(s) in the home-cage of the animals. In the delayed spatial alternation task (Figs.2), mice (n = 6) were trained to run in the figure-8 maze, starting at the beginning of the center stem and having to alternate between left and right arms of the maze in each trial. An 8-s delay was introduced at the beginning of each trial, during which the animal was confined at the center stem by automatic doors. Upon completion of a correct trial, a small reward (30% sugar water) was provided at the end of the side arms. Maze doors and water delivery were fully automated with a custom circuit. Animals ran two or three

behavioral sessions per day (~20-40 trials for ~30 mins per session), interleaved with 2-hour sleep sessions.

In the novel versus familiar task (**Extended Data**. 2a), mice (n = 5 for M1 and M2were first trained on the delayed alternation task on one maze (M1, 61-cm diameter), as described above. After mice consistently surpassed 80% correct trials per session (final performance = 83.9 ± 3.2%, as mean ± sem for 5 mice), which took 4-5 days, the second figure-8 maze (M2) was introduced. These two mazes had different spatial layouts (Fig.1c), local cues (textures, walls and landmarks) and were located in different rig enclosures with distinct distal cues. On each day, mice ran in M1 and M2 in morning and afternoon sessions (in a variable order) separated by a 2-hour rest period in their home cage.

Rat behavioral tasks:

Before surgery, animals were accustomed to the maze for at least one week. After surgery recovery, rats were handled daily and accommodated to the experimenter, recording room, and cables for at least a week before the start of the experiments. One day before the beginning of the training, rats were placed under water restriction. All recording sessions included a session of sleep before and another session after the performance of the task(s) in the home-cage of the animals.

In the cheeseboard task, rats were trained to find three water rewards hidden in the fixed locations of the platform (120 cm diameter) during several trials (up to 30). The location of the rewards was changed per day, to involve a new spatial learning in each recording session. During the first five trials of the day, animals were exposed to the same reward configuration as the previous day (C0), to test for the "recall" of the previous day locations. After this, 20-30 trials of new spatial learning (i.e., new reward's configuration, C1) followed, with sleep sessions of ~1.5-2 hours preceding (pre-sleep) and following (post-sleep) the new learning trials. During the new learning, a given trial was considered completed once the rat had collected the three rewards and returned to the start-box to collect an additional food pellet. A period of ~30-60 seconds was imposed in the start-box to allow experimenters to refill reward locations. A deadline of 2 minutes was imposed, after which animals were redirected to the start-box if they had not yet collected all three rewards. After the post-sleep following the new learning, animals were exposed to either a) a second spatial configuration of the rewards (i.e., a second novel configuration, C2) or b) the same novel configuration that was recently presented (C1'). Learning performance was estimated as either 1) the latency from the start-box to collect all three rewards, 2) the distance traveled from the start-box to collect all three rewards (quantified as latency or distance traveled to collect the three rewards divided by a session-specific optimal trajectory) or 3) the number of trials to collect all three rewards (measured as the first time the path length was lower than mean + 1 SD of the optimal trajectory). For each session, we computed the cumulative distance (or cumulative time) for each trial, and the average across the last N/2 trials is used as the optimal baseline trajectory. A value of 1 indicates an optimal path taken by the animal. Individual trial performance was normalized by this baseline (normalized path length or normalized time). The early learning performance is measured by the average normalized path length or normalized time over the first 5 trials, z-scored within each animal. The d_2 as well as other variables including speed, firing rate and theta during learning and recall were computed across the concatenated first five trials from each epoch. Periods of maze running

were considered from when an animal entered the maze until it collected all three rewards. The d_2 during inter-trial intervals was computed considering the 15s before an animal ran into the maze.

A performance threshold was set to mean + 1SD of the latter-half trials, and a measure of learning speed (acquisition trials) was defined as the first trial at which the trial cumulative distance fell below that threshold, then z-scored within each animal. For example, in a 30-trial session the baseline and threshold were computed from trials 16–30. Performance in the test sessions was computed in the same way. The inclusion of data for the analysis required that the animal be pretrained for a week in this task.
In all the behavioral sessions, animals' position was tracked with an overhead camera (Basler acA720-520uc and acA1300-200uc), and later extracted using DeepLabCut (DLC, v2.3.0)[83].

Epileptic model

Epilepsy was induced following previously described methods[84,85]. Rats received intraperitoneal injections of kainic acid (5 mg/kg) at hourly intervals until they developed *status epilepticus*. Seizure periods were identified using the Racine scale [86]. Status epilepticus was defined as a continuous hyperactivity state in LFP lasting more than 10 minutes.

Recording system and data preprocessing

An Intan RHD2000 interface board or Intan Recording Controller was used for electrophysiological recordings. The sampling rate was set at 20 kHz. Both amplification and digitization were done in the head stage (Intan Technologies). Data were visualized online during recording using the Intan software and Neuroscope (Neurosuite). LFP signals were down-sampled at 1250 Hz for subsequent analysis.

Tissue processing and immunohistochemistry

After experiments were terminated, mice were deeply anesthetized and kept under high-flow (4-5%) isoflurane anesthesia. They were then perfused transcardially with 0.9% phosphate buffer saline (PBS) solution followed by 4% paraformaldehyde (PFA) solution. Brains were kept in PFA for 24 hours and if necessary, in PBS afterwards until further processing. Brains were then sectioned into 70-µm thick slices (Leica Vibratome, 2000). Finally, sections were washed and mounted on glass slides with fluorescence medium (Fluoroshield with DAPI – F6057, Sigma, USA). Immunostained slices were examined, and histological images were acquired with a Zeiss confocal microscope.

Quantification and statistical analyses

Spike sorting and single unit classification

Spike sorting was performed semi-automatically using KiloSort [87] (https://github.com/cortex-lab/KiloSort), followed by manual curation using the software Phy (https://github.com/kwikteam/phy) and custom designed plugins (https://github.com/petersenpeter/phy-plugins) to obtain well-isolated single units. Cluster quality was assessed by manual inspection of waveforms and auto-correlograms, and by isolation distance metrics. Multi-units, noise clusters, or poorly isolated units were discarded from further analysis. Well-isolated units were classified into putative cell types using the MATLAB package, Cell Explorer [88] (https://github.com/petersenpeter/CellExplorer). Spiking features such as auto-correlogram (ACG), spike waveform, and putative monosynaptic connections derived from short-term cross-correlograms (CCGs), were used to characterize and classify well-isolated units [81,89,90]. Three cell types were assigned: putative pyramidal cells, narrow waveform interneurons, and wide waveform interneurons. The two key metrics used for this separation were burst index and trough-to-peak latency. Burst index was determined by calculating the average number of spikes in the 3-5 ms bins of the spike ACG divided by the average number of spikes in the 200-300 ms bins. To calculate the trough-to-peak latency, the average waveforms were taken from the recording site with the maximum amplitude for the averaged waveforms of a given unit. Only putative well isolated neurons were used for further analysis, unless otherwise specified, and all sessions were included in the analyses unless otherwise specified.

Sleep scoring and brain state classification

Broadband LFP (i.e., slope of power spectral density), narrow-band theta frequency LFP, and estimated electromyogram (EMG) were used as input into the algorithm for estimating brain states as previously described [91,92]. Briefly, spectrograms were computed from broadband LFPs using a fast Fourier transform with 10 s and sliding windows of 1 s. PCA were then computed after a *Z*-transform. The first component reflected power in the low (< 20Hz) frequency range, with oppositely weighted power at higher (> 32 Hz) frequencies. Theta dominance was quantified as the ratio of powers in the 5-10 Hz and 2-16 Hz frequency bands. EMG was estimated as the zero-lag correlation between 300-600 Hz filtered signals between recording sites. Soft sticky thresholds on these metrics were used to identify sleep states. High LFP principal component and the low EMG were considered non-rapid eye movement (NREM) sleep, the high theta and low EMG were considered REM sleep, and the remaining times were taken as WAKE. To avoid spurious transition detection due to noise, a "sticky" threshold was used [92]. Accelerometer fluctuations (i.e., a signal that does not depend on LFPs) extracted from the preamplifier (Analog Devices ADXL335 3-axis accelerometer) were used as an alternative estimate of movement to validate analyses of neural and pupil dynamics correlation with EMG.

Criticality analysis

Detection of neuronal avalanches

Neuronal avalanches were detected as in previous studies [8,25,28,45]. Briefly, within a given session or time interval (i.e., pre-, mid- or post-sleep), spike trains were binned into 40 ms bins, and a total spike count (from the activity of all neurons or multiunit activity -MUA) was computed for each time bin. A threshold (30th percentile for mice) was applied to define active bins. A given avalanche beginning and end were set as the time stamps of when MUA passed above and fell below the threshold respectively.

The distributions of both avalanche size (S) and duration (T) were fitted with a power-law model using the *powerlaw* python package [93,94], as $P(S) \sim S^{-\tau}$ and $P(T) \sim S^{-\alpha}$. For each condition, the model parameters were obtained using a maximum likelihood estimation (MLE) on the probability distribution function (PDF), with user-defined lower and upper truncation bounds. The exponents of the distributions were estimated directly from the empirical data, and the *goodness of fit* was evaluated using the Kolmogorov–Smirnov (KS) statistics between the empirical distribution and the fitted power-law model. The exponent of the distribution of avalanche size (tau) and duration (alpha) are shown in (Extended Data Fig.2 e, f).

For visualization purposes only, empirical complementary cumulative distribution functions (CCDFs) were computed and plotted on a log–log axis together with the theoretical CCDF derived from the fitted PDF, scaled to match the empirical CCDF at the minimum value (x_{min}).

Scaling ratio and deviation from criticality (DCC)

The mean avalanche size conditioned on duration exhibits a power law given by $\langle S \rangle \langle T \rangle \sim T^{\beta}$ under the critical regime according to the crackling noise scaling law, where $\beta = \frac{1-\alpha}{1-\tau}$ [28,45,95,96]. The empirical scaling ratio was obtained by fitting a linear regression to the probability density function (PDF) of avalanche size ⟨S⟩ and duration ⟨T⟩ in a log–log space. The absolute difference between the empirical and theoretical scaling ratios (exponents of the PDF of avalanche size and duration, alpha and tau, respectively) was defined as the deviation from criticality (DCC) [25,45].

Branching ratio

The branching ratio was estimated using a previously developed method based on the multistep regression (MR) method implemented in the mr*estimator* toolbox [24,97], which provides an unbiased estimate of intrinsic branching dynamics independent of subsampling. Briefly, population spike counts were first binned in 4 ms bins, capturing the typical propagation time of spikes between neurons. The branching ratio was then computed within sliding windows of 10 s with a step size of 2 s. For each time lag *k*, a linear regression was performed between A_t and A_{t+k}across all time points and sessions as in:

$$A_{t+k} = r_k A_t + c_k \#(1)$$

where A_t representing the population spike count at time bin *t*, yielding a set of lag-dependent slopes $\{r_k\}$. Each lag *k* yields a regression coefficient $\{r_k$, with r_1 corresponding to the conventional (but subsampling-biased) estimator of the branching ratio *m*. Under subsampling, all regression slopes r_k are biased by the same multiplicative factor. Consequently, instead of the ideal full sampling relation, the regression coefficients follow the generalized form:

$$r_k = b \cdot m^k \#(2)$$

where *b* is a constant scaling factor reflecting the degree of subsampling. The branching ratio *m* was then estimated by fitting this exponential relationship across all lags *k* using nonlinear least-squares optimization.

Analyses were done separately for different brain states unless otherwise specified. To that end, for each brain state (NREM, REM, wake-theta or wake non-theta) obtained with the state-scoring classification (see Sleep scoring and brain state classification section), a state-specific temporal mask was applied to ensure that the branching ratio captured the temporal dynamics of a specific state. The change in branching ratio (ΔBR) around either SWRs or BARRs events was computed using values within the central window (−2.5 to +2.5 s) relative to baseline values (considered as −20 to −18.5 s for pre-event windows and +18.5 to +20 s for post-event windows).

Temporally resolved distance to criticality estimation

The temporally resolved distance to criticality was calculated using the temporal renormalization group (tRG) framework, using autoregressive (AR) models to provide a scalar measure of the proximity to criticality for each window of a time series (Sooter *et al.*, 2024). Briefly, population spike counts (multiunit activity -MUA) were binned into 40 ms bins and z-scored to obtain a model of neural activity (x_t) at time step *t,* fitting an AR(p) model (order p = 20) using the Yule-Walker method as:

$$x_t = \sum_{i=1}^{p} \phi_i x_{t-i} + \xi_t \, , \xi_t \sim N(0, \sigma^2) \#(3)$$

Where ξt is gaussian noise with variance 2, ϕ_i represents the coefficients of the history kernel in recent history (p time steps) that influence x_t, and determines whether a given model AR(p) will fall into a fixed point. Analyses were performed using sliding windows of 30 s or 60 s, with step sizes of 2 s and 5 s, respectively. For each AR model, the exponent of the low frequency part of its power spectrum is defined as $-\beta$. With coarse graining, a family of AR models will flow into fixed points of the hyperplane given by β=0 (white noise fixed point), β=2, β=4… The Euclidean distance from the AR model to each hyperplane (β) define de distance, d_β, to criticality. In our analyses, the coefficients of the AR model that best fit the history kernel of the empirical data were used to compute the distance to the β = 2 hyperplane [44], thus, we refer to the distance to criticality as d_2:

$$d_2 = \|\widehat{\phi} - \phi\| \#(4)$$

Given this, d_2=0 for an AR model that captures temporal correlations of criticality, and increases smoothly as the dynamics of the time series represented by the model deviate from the critical manifold β = 2. Analyses were done separately for different brain states unless otherwise specified. To that end, for each brain state (NREM, REM, or Wake) obtained with the state-scoring classification (see Sleep scoring and brain state classification section), a state-specific mask temporal mask was applied to ensure that d_2 captured the temporal dynamics of a specific state.

Estimation of uncertainty in Euclidean distance

The uncertainty in d2 was estimated using the delta method adopted from [44], by propagating the full Hessian-based covariance of the AR coefficients onto the Euclidean distance metric (onto

the Euclidean space used to calculate the distance to criticality). Specifically, the covariance was approximated as $Cov(\phi) \approx \sigma^2(X^\top X)^{-1}$, and the standard error of d_2 was computed as:

$$\Delta d_2 = \sqrt{g^\top Cov(\phi) g}, g = \frac{\hat{\phi} - \phi_2^*}{\| \hat{\phi} - \phi_2^* \|_2} \#(5)$$

Sharp-wave ripples, replay and barrages analyses

Event-rate sextiles were computed using the same 30 s time bins and 2 s step size as the d_2 analysis. Within each session, event rates were first partitioned into six equal-width bins (or quantiles), and the resulting values were then pooled across all animals for group-level analysis.

Sharp-wave ripples

For SWR detection, the wide-band signal from a CA1 pyramidal layer channel was filtered (difference-of-Gaussians; zero-lag, linear phase FIR), and instantaneous power was calculated by clipping at 4 SDs, rectified, and lowpass filtered [47,53,80,81,98]. The low-pass filter cut-off was at a frequency corresponding to p cycles of the mean bandpass (for 80-250 Hz band-pass, the low-pass was 55 Hz). Subsequently, the power of the non-clipped signal was computed, and all events exceeding 4 SDs from the mean were detected. The events were then expanded until the non-clipped power fell below 1 SD; Sharp waves were detected separately using LFP from a CA1 str. radiatum channel, filtered with band-pass filter boundaries (5-40 Hz). LFP events of a minimum duration of 20 ms and a maximum of 400 ms exceeding 2.5 SD of the background signal were included as candidate SWRs.

Replay analysis

Detection of replay sequences during SWRs was based on previously described methods [47,52,59,99] (two out five T-maze mice without high-density probes were excluded from this analysis). In brief, the high synchrony events (HSEs) during immobility period (less than 1 cm s^{-1}) were first detected as previously reported[100]. To detect HSEs, the combined activity of all pyramidal spikes was binned into 1-ms bins and smoothed with a Gaussian kernel (s.d. = 15 ms) to produce a population firing rate curve over time. HSEs were initially detected as contiguous periods when the population firing rate stayed above 3 s.d. of the mean, and further refined as times around the initially detected events during which the firing rate exceeded the mean. Only HSEs with a duration of 50 ms or more were included for further replay analysis. Candidate replay events were defined as the HSEs during which five or more place cells fired and overlapped with at least one SWR[52]. Each candidate event was then divided into 10-ms non-overlapping bins. For each time bin, a memoryless Bayesian decoder was built for the two trajectory types in a T-maze to estimate the probability of the position of the mice given the observed spikes (Bayesian reconstruction; or posterior probability matrix):

$$P(X, Tr|spikes) = \frac{P(spikes|X, Tr)P(X, Tr)}{P(spikes)} \#(6)$$

assuming independent Poisson firing and a uniform prior. Replay structure was quantified using weighted correlation of the posterior probability matrix. Statistical significance was assessed via

circular time-bin shuffling (n = 1,000); events exceeding the 97.5th percentile (or below the 2.5th percentile for reverse replay; $p < 0.025$) were classified as significant. The trajectory with the lowest p-value was assigned as the replayed trajectory, with replay probability defined as 1-p. Replay probability was defined as the number of ripples with significant replay divided by the total number of ripples in that window.

Barrage analysis

BARRs were detected using the single unit activity within a subpopulation of CA2 pyramidal cells as previous report [53]. First, a threshold was used for extracting the cells that can sustain burst-like spiking persistently (frequency higher than 30 Hz during more than 300 ms). Using this subpopulation of CA2 pyramidal cells, a binned (5 ms per bin) spike-train was constructed and convolved with a Gaussian kernel (20 ms). A candidate event was defined if a given threshold (between 3 to 5 standard deviations) over the mean was crossed for a minimum of 150 to 200 ms during NREM epochs. BARR events were further restricted to requiring between 2-3 units to fire 4-7 spikes in an event.

Background d_2 with ripple ablation

For NREM bouts, windows were further "ripple-ablated" by subtracting ripple intervals that overlapped into that period); this produced one or more ripple-free subwindows per NREM bout. Windows in which the total remaining ripple-free duration of NREM was < 1 s were discarded. For the remaining windows, pooled spike times were binned at t=40ms and analyzed, together with the ripple-free subwindows, using the AR/tRG routine described above. The resulting d spectrum was computed for each window, and the value at $\beta=2$ was taken as the ripple-ablated distance to criticality d_2, with its uncertainty obtained from the Hessian–delta-method, as described above.

Inter-spike-interval entropy

Inter-spike-interval (ISI) entropy was calculated to estimate the entropy of the firing dynamics during different windows of time. Unlike firing-rate-based entropy which collapses the temporal spike structure into a single count, ISI entropy directly quantifies the variability of spike timing and is independent of mean firing rate when computed in a logarithmic time, yielding more accurate and less biased estimates of neural information [101].
For each analysis window, we quantified firing irregularity using the Shannon entropy of the ISI distribution in log-time, based on previously developed methods used for information theory analyses of neuronal spike trains [101–103]. Three spikes were required for calculating the entropy in a given window, otherwise the ISI entropy was not defined for that window. The ISI was computed as:

$$\Delta t_j = t_{j+1} - t_j \#(7)$$

Where t_k indicates the spike times (in seconds) within a given window, pooled and sorted after removing duplicates, negative and non-finite values. We then transformed it to a logarithmic scale using:

$$x_j = (\Delta t_j) \#(8)$$

and estimated the empirical distribution of x_j using a fixed histogram 20 equally spaced bins such:

$$N = \sum_i \quad n_i \#(9)$$

where n_i is the count in bin i and N representing the total number of ISIs contributing to the histogram. Note that the entropy was not defined when N=0. The corresponding probabilities were calculated as:

$$p_i = n_i/N \#(10)$$

with empty bins resulting in $p_i = 0$. The final ISI entropy was defined as the Shannon entropy in bits as:

$$H_{ISI} = -\sum_i \quad p_i(p_i) \#(11)$$

which yielded a single scalar H_{ISI} per window (in bits), with higher values indicating a broader, more irregular ISI distribution in log-time.

Ripple rate and MUA versus d_2 correlations

Ripple rate, multiunit activity (MUA) and d_2 were first *z*-scored across periods of interest within a session. Spearman or Pearson correlation coefficients were calculated between each feature and the corresponding variable to quantify either monotonic (Spearman) or linear (Pearson) trends.

Multiunit firing dynamic range

Multiunit activity (MUA) from pyramidal cells and interneurons was binned in 40-ms windows, then smoothed by averaging within a 30-s sliding window advanced in 2-s steps. The dynamic range was then calculated as the logarithmic ratio between the 95th and 5th percentile of multiunit firing [2] in a given bin.

Generalized linear model

To quantify state-dependent predictors of replay, we used a regularized logistic regression to model replay occurrence (binary) at the event level as described in previous work [104,105]. Events were assigned to one of six ordered state bins derived by quantile binning of the state variable (e.g., d_2), preserving rank order while equalizing sample counts across bins. Continuous predictors (d_2, MUA, FR_std) were z-scored before modeling. We compared a reduced model containing only main effects (state bin and predictors) with a full model that additionally included feature × state interaction terms, allowing predictor effects to vary across state bins.

Models were evaluated using 5-fold cross-validation, grouped by session when available to prevent data leakage. Within each fold, predictors were standardized using training data only. Predictive performance was assessed using the area under the Receiver Operating Characteristic curve (AUC). State-dependent predictive gain was quantified as the improvement in out-of-fold performance of the full model relative to the reduced model [106]. To assess the contribution of individual predictors to state-dependent gating, interaction terms were ablated one at a time from the full model, and the resulting change in performance was measured. Statistical significance of predictive gain was then evaluated using nonparametric tests against zero. In this way, positive predictive gain indicates that the removed interaction term improves prediction and is necessary for capturing state-dependent gating.

Place field analysis

Spiking data and the tracked animal's position were binned into 3 cm wide segments of the maze floor, generating raw maps of spike counts and occupancy for cells with at least 100 spikes in one of the learning sessions. A Gaussian kernel (SD = 3 cm) was applied to both raw maps of spike and occupancy, and a smoothed rate map was constructed by dividing the spike map by the occupancy map. Independent rate maps were constructed for the different task learning epochs in the mazes (Fig.4). Only periods in which the animal velocity was greater than 4 cm/s were included.

Neural Latent Representations

To characterize the low-dimensional latent representation, we computed population-vector embeddings using self-supervised time-contrastive learning implemented with CEBRA-time [70]. CEBRA-time discovers the underlying manifold by optimizing a neural network to distinguish between temporally proximal "positive" pairs $\{z^{+}_{i}\}$ and randomly sampled "negative" pairs $\{z^{-}_{i,j}\}$ for any anchor z_i, via a variation of the InfoNCE loss function [107]:

$$L = -\sum_{i} \; log \; \frac{exp(s(z_i, z_i^{+})/\tau)}{exp(s(z_i, z_i^{+})/\tau) + \sum_{j} \; exp\big(s\big(z_i, z_{i,j}^{-}\big)/\tau\big)} \; \#(12)$$

where s denotes cosine similarity and τ is a temperature hyperparameter. By enforcing the contrastive objective, CEBRA-time ensures that states close in time are mapped to nearby points on a latent manifold, effectively uncovering the intrinsic geometry of the neural dynamics without the need for explicit behavioral labels or supervision[70].
Latent population embeddings were extracted from CA1 neurons (n = 111.13±46.15 cells per session, 30 sessions, 3 animals). Spikes from all pyramidal neurons were binned into 50 ms windows and normalized within each session. Recordings were parsed into trials (one round trip from the start box per trial); a 15 s period prior to the animal leaving the start box was also included. A speed filter (> 3 cm/s) removed low-velocity samples to avoid overrepresentation of reward sites. A trial-level shuffle permuted the order of continuous trials while preserving

within-trial temporal structure, and time permutation shuffling was optionally applied to estimate chance performance. Low dimensional embeddings were learned with the CEBRA-time model trained on the trial-shuffled data (model architecture: 'offset10-model'; batch size: 128; learning rate: 3×10^{-4}; temperature: 1; distance: 'cosine'; time offsets: 8; max iterations: 3×10^4; conditional: 'time'; output dimension: 10).

Separability analysis in latent space

A support-vector classifier (SVC) was trained to decode task configuration from the latent embeddings. Hyperparameter selection was performed with GridSearchCV over the grid $C \in \{0.1, 1, 10\}$, $\gamma \in \{$'scale', 'auto', 0.01$\}$ with kernel set to 'rbf', using 5-fold cross-validation. For each run we recorded the best cross-validated accuracy. Session level separability was computed as the mean accuracy of the best SVC across five independent repetitions of the training and model-selection procedure. Label shuffling was optionally applied to estimate chance performance.

Cross regional communication analysis

To quantify cross-regional communication between CA1 and its input regions (CA2, CA3, MEC) we used Canonical Correlation Analysis (CCA) to measure shared population activity structure. Briefly, for each session and region-pair (e.g. CA3 vs CA1) spike trains for each neuron were binned into 20 ms non-overlapping windows and smoothed with a gaussian kernel with SD = 40 ms to obtain binned firing-rate time series. For each neuron the binned firing-rate time series was z-scored (zero mean, unit variance) across the analyzed epoch to remove firing-rate scale differences. For any pair of regions, $X \in R^{T \times nX}$ and $Y \in R^{T \times nY}$ be the z-scored population matrices (T time bins, nX and nY neurons). CCA finds weight vectors a and b that maximize the correlation:

$$\rho = corr(Xa, Yb) \#(13)$$

We compute the full set of canonical correlations $\{\rho_1 \geq \rho_2 \geq \ldots\}$. As our summary measure of population coupling we report the first canonical correlation ρ_1, which captures the strongest linear relationship between the two population activity patterns [108]. To test session-level changes (e.g. beginning vs end of learning) we used the Wilcoxon signed-rank test across sessions on the empirical popCorr values. Neuron numbers larger than 60 were randomly resampled to 60 to avoid bias from unequal neuron counts (10 times) and average population correlation is reported.

CA1 network model

The simplified CA1 network model consisted of 8000 pyramidal cells (E), 200 parvalbumin inhibitory interneurons (PV) and 200 CCK interneurons simulated using the Brian2 simulation environment [63,109]. Input from CA3 was modeled as Poisson spike trains with a mean firing rate variates in [1, 28] Hz, projecting to all three populations and the basic synaptic weights are tuned to generate a balanced state in the network. Input from CA2 is modeled as Poisson spike trains with a mean firing rate variates in [0, 25] Hz, projecting to each CCK neuron and the synaptic weights are tuned to generate a critical state in the network. All cell types are modeled as conductance-based adaptive exponential integrate-and-fire (AdExpIF) cells [110] following:

$$C_m \frac{dV}{dt} = -g_L(V - E_L) + g_L \Delta_T e^{(V - V_T)/\Delta_T} - w + I_{syn}(V, t) \#(14)$$

where C_m is the membrane capacitance, g_L is the leak conductance, E_L is the resting potential, Δ_T is the slope factor, V_T is the spike threshold, w is the adaptation current, I_{syn} is the total synaptic current. The synaptic current is given by:

$$I_{syn}(V, t) = \sum_i \quad g_i(t)\big(V - E_{syn_i}\big) \#(15)$$

where $g_i(t)$ is the synaptic conductance and E_{syni} is the reversal potential of the synapses (i.e. 0 mV for excitatory synapses and −70 mV for inhibitory synapses). The synaptic conductance follows a biexponential kinetics [111]:

$$g_i(t) = g_{max} A\big(e^{-(t - t_{spike})/\tau_d} - e^{-(t - t_{spike})/\tau_r}\big) \#(16)$$

where g_{max} is the maximum conductance, τ_d and τ_r are the decay and rise time constants, and t_{spike} is the time of the presynaptic spike. The normalization constant $A = exp(-1/\tau_d) - exp(-1/\tau_r)$ was chosen such that the synapses reach their peak conductance at $t_p = (\tau_d \tau_r)/(\tau_d - \tau_r)\ ln\ ln\ (\tau_d/\tau_r)$ ms.

The adaptation current w in equation .14 follows:

$$\tau_w \frac{dw}{dx} = a(V\ -\ E_L) - w \#(17)$$

where a is the subthreshold adaptation and τ_w is the adaptation time constant. When the membrane potential V reaches the spike threshold V_{spike}, a spike is emitted, the membrane potential is reset to V_{reset} for a refractory period of τ_{ref}, and the adaptation current is increased by a spike-triggered adaptation b. We used previously described kinetic parameters from hippocampal recordings summarized in the table below [62,63,65,109]. Synaptic connectivity between hippocampal populations was modeled as a random sparse connectivity with connection probability p and synaptic weights w_{syn} for each connection type (E-E, E-PV, PV-E, PV-PV, PV-CCK, CCK-E, CCK-PV, CCKCCK). In line with experimental results [64,65], E-CCK connectivity was not included in the model. Similarly, PV-PV connectivity is not included for CA1 network.

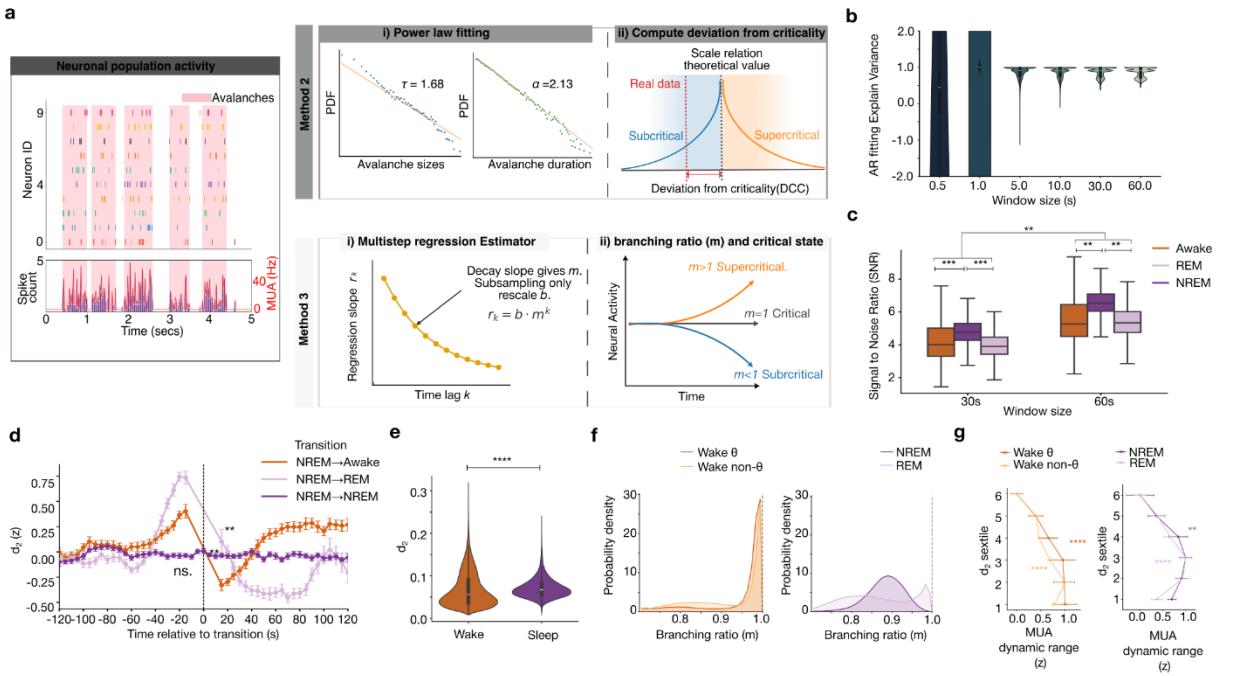
Simulations were run for 60s with a timestep of 0.1ms. MUA was then computed by summing all spikes from pyramidal neurons in 5 ms bins (avrage ISI = 0.18 ms) [43]. Distance to criticality d_2 was estimated from the population activity of pyramidal neurons in sliding windows of 10s with 5s overlap. All the simulations were repeated for n = 10 sessions with different random initial conditions.

	C_m (pF)	g_L (ns)	E_L (mV)	V_T (mV)	Δ_T (mV)	V_{reset} (mV)	t_{ref} (ms)	τ_ω (ms)	a (nS)	b (pA)
E	180.13	4.31	-75.19	-24.42	4.23	-29.74	5.96	84.93	-0.27	206.84
PVBC	118.52	7.51	-74.74	-57.71	4.58	-64.99	1.15	178.58	3.05	0.91
CCKBC	73.63	3.84	-64.27	-44.95	4.88	-52.1	2.00	152.10	0.81	25.90

Statistical Analyses

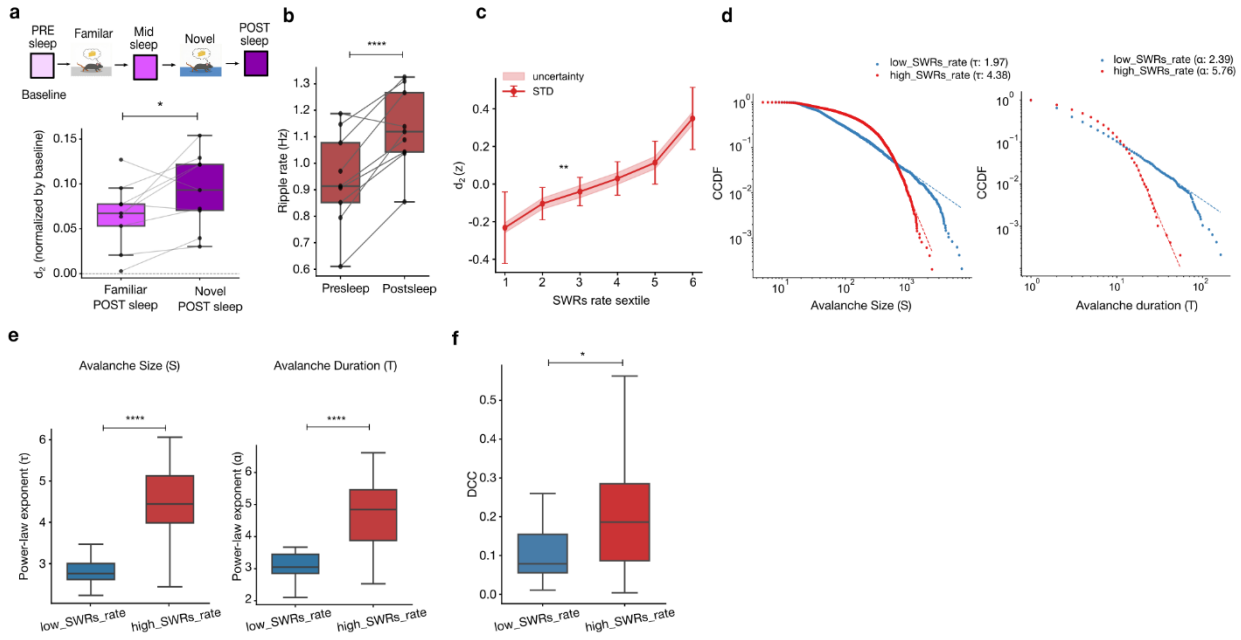
Data analysis was performed using custom routines in Python or MATLAB (MathWorks). No specific analysis was used to estimate minimal population sample or group size, but the number of animals, sessions and recorded cells were larger or similar to those employed in previous related work [47,52,53,59,67,80–82]. Unless otherwise noted, non-parametric Wilcoxon rank-sum or Wilcoxon signed-rank test Linear mixed-effects models (LMMs) were used for paired and unpaired comparisons respectively, with random intercepts for animal and session nested within animal. [112,113]. All statistical tests were two-tailed except when indicated specifically with $p < 0.05$ as the cutoff for statistical significance, which is indicated by asterisks (*p<0.05, **p<0.01, ***p<0.001, and ****p<0.0001) with values exceeding numerical precision reported as $p < 10^{-15}$. Error bars show standard error mean (SEM), and boxplots show median (central mark), 75th (box), and 90th (whiskers) percentile, unless indicated otherwise. For boxplots, points below the 10th percentile and above the 90th percentile are defined as outliers; outliers are not displayed in the plots but included in statistical analysis. Due to experimental constraints of optogenetic experiments, experimenters were not blind to these manipulations.

Extended Data Fig. 1 to 10

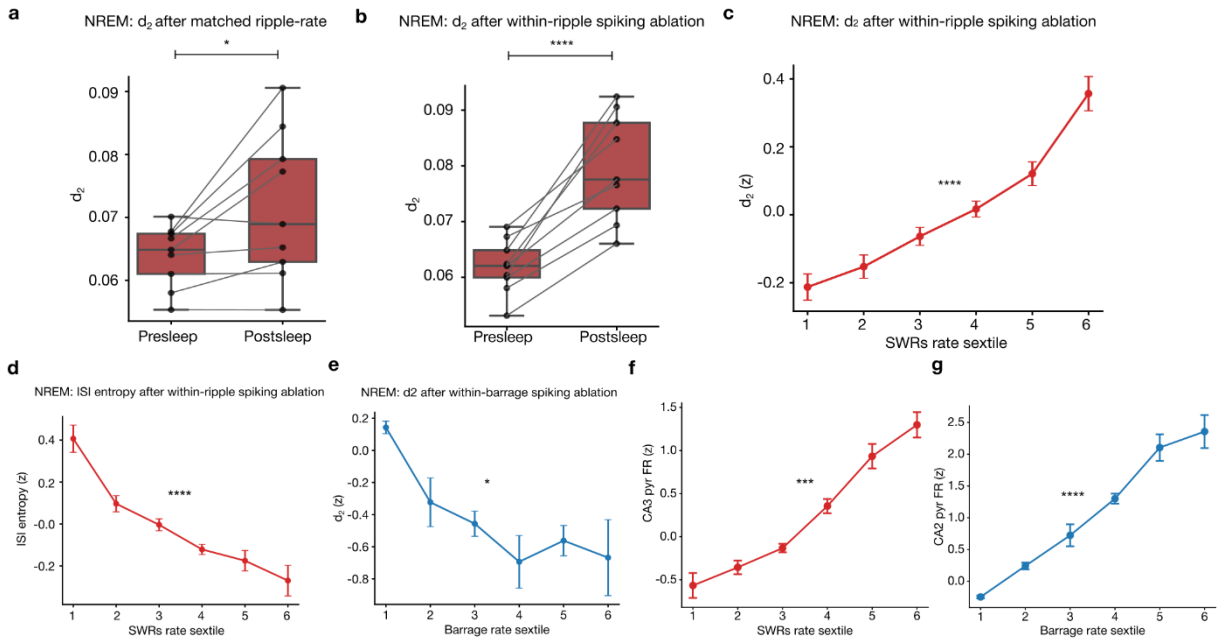


Extended Data Fig.1 a, Left**:** schematic of avalanche detection over simulated spike trains. Top (method 2): workflow for criticality measures based on power law fitting. i) Example of power law distribution of the size (blue) and duration (green) of neuronal avalanches and associated linear fits (orange lines). ii) schematic of quantification of deviation from criticality as the difference between the exponents of the theoretical and the empirical values of the fitted slope. Bottom (method 3): workflow for criticality measures based on branching ratio. i) Schematic of the multistep regression estimator (Wilting–Priesemann–type) used to calculate the slope of the exponential decay across population activities separated by a time lag k, allowing the recovery of the intrinsic branching ratio (*m)* independent of subsampling. ii) schematic of estimation of distance to criticality based on the value of the branching ratio (*m)*, with subcritical ($m<1$), critical ($m=1$), and supercritical ($m>1$) regimes indicated in blue, black and orange respectively. **b,** Explained variance from autoregressive (AR) model fitting to population activity for sliding windows of different durations. Estimates were run over a random subset of the dataset (n=6 sessions from 4 animals). **c,** Distributions of signal-to-noise ratio (SNR) across 30-s and 60-s windows for AWAKE, NREM, and REM epochs across all sessions. SNR during NREM was higher than during WAKE for 30s (n = 18 sessions from n = 6 animals, $p = 1.53\times10^{-4}$) and 60s windows (n = 11 sessions from n = 6 animals, $p = 2.93\times10^{-3}$) and higher than during REM for 30s (n = 16 sessions from n = 6 animals, $p = 1.53\times10^{-4}$) and 60s windows (n = 9 sessions from n = 6 animals, $p = 7.81\times10^{-3}$). SNR for 60s windows was higher than for 30s windows when comparing across states (n = 11 sessions from n = 6 animals, $p = 2.93\times10^{-3}$). Pairwise comparisons with Holm-Bonferroni correction were used for all comparisons. Note that states lasting <40 s (for 30-s windows) or <70 s (for 60-s windows) were excluded from the analyses. **d,** Mean time-resolved d_2 surrounding state transitions (transitions are centered at t = 0). NREM→Awake, NREM→REM, and NREM→NREM transitions are shown in orange, pink

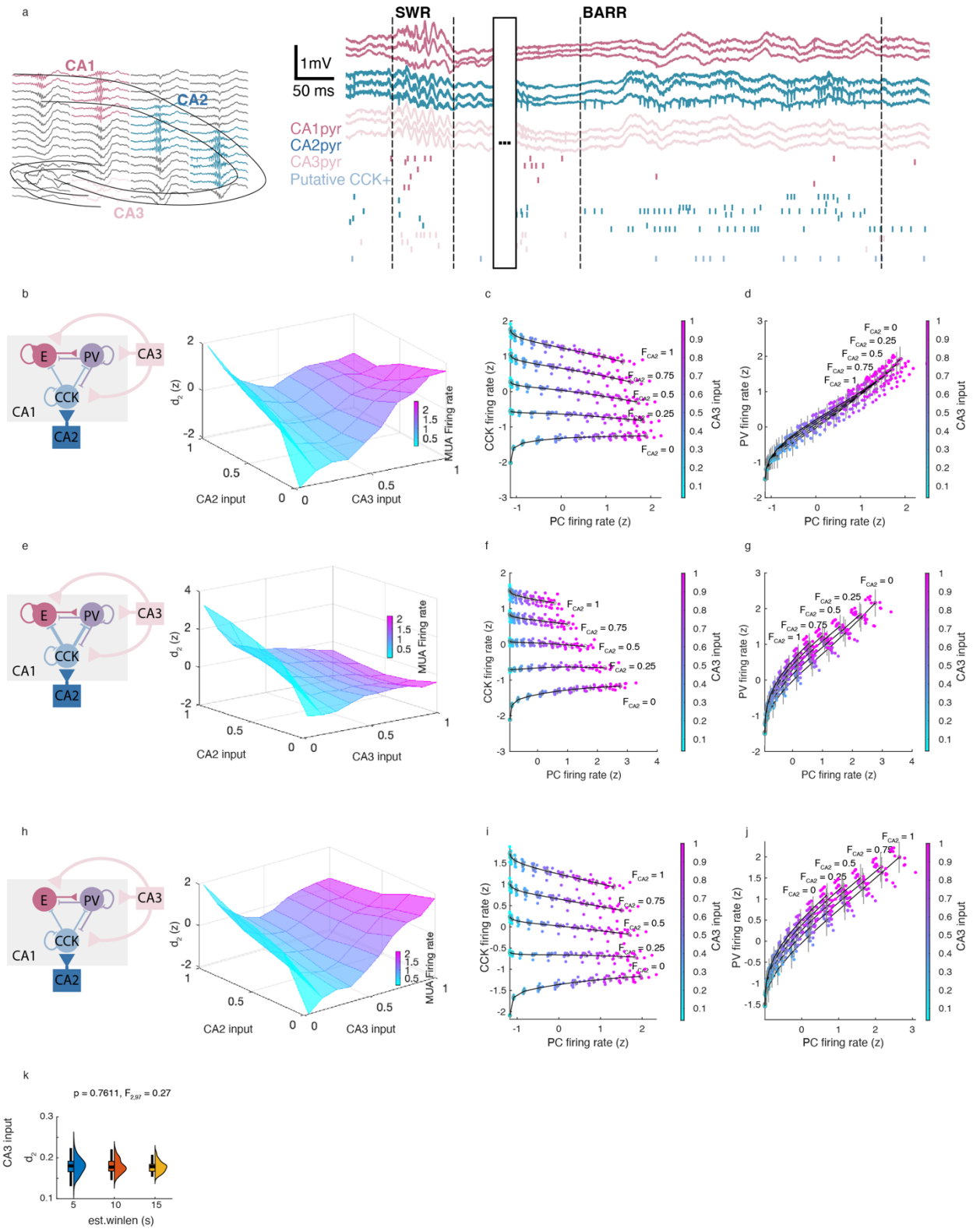
and purple respectively. A negative trend was found for NREM→Awake transitions (Spearman's: $\rho = -0.18$, $p = 0.0023$) and NREM→REM transitions (Spearman's: $\rho = -0.44$, $p = 0.0029$). As a control, NREM→NREM transitions do not exhibit significant changes ($p > 0.95$). n = 11 sessions from n = 6 animals were included for all comparisons. **e,** d_2 distributions for wake (orange) and sleep (purple) ($p = 1.58 \times 10^{-6}$; paired Linear Mixed Model, n = 18 sessions from n = 6 animals). **f,** Branching ratio (*m*) distributions for theta and non-theta states during wake (left panel) and REM and NREM stages during sleep (right panel). **g,** Distance to criticality, d_2, in relation to multiunit firing dynamic range (see Methods). Note that a higher dynamic range corresponds with closer distance to criticality (lower d_2) ($\rho = -0.34$, $p = 3.97 \times 10^{-3}$ and $\rho = -0.53$, $p = 8.95 \times 10^{-5}$ for NREM and REM respectively, and $\rho = -0.80$, $p = 3.22 \times 10^{-11}$ and $\rho = -0.84$, $p < 10^{-15}$ for awake θ and awake non-θ respectively (One-sided Spearman correlation, n = 18 sessions from n = 6 animals).



Extended Data Fig.2 a, d_2 during post-familiar task sleep (F) or post-novel task sleep (N) normalized to baseline (pre-task) sleep. Animals typically undergo presleep →familiar task → midsleep→F/N task→postsleep. d_2 during post-novel task sleep was higher than post-familiar task sleep (p = 0.030, paired Linear Mixed Model, n = 9 sessions from n = 5 animals). **b,** SWRs rate was higher during post-task compared to pre-task NREM sleep ($p=2.98\times10^{-8}$, paired Linear Mixed Model, n=9 from n = 6 animals). **c,** Mean d_2 (z-scored) increased monotonically with ripple rate (linear regression, OLS: $R^2 = 0.94$, $p = 0.001$; linear regression weighted by STD: $R^2 = 0.94$, $p = 0.002$; linear regression weighted by propagated error: $R^2 = 0.95$, $p = 0.001$). Error bars denote standard deviation, and shaded regions represent propagated uncertainty of the d_2 estimator [44]. **d,** Complementary cumulative distribution functions (CCDFs) of avalanche size (left) and avalanche duration (right) during periods of low SWRs rate (blue) and high SWRs rate (red) from a representative session. Dashed lines indicate linear fit of the power-law estimated by maximum likelihood, and α values indicate exponents of the linear fit. Note that during periods of high SWRs rate, the CCDF of avalanche size and duration decayed more steeply. **e,** Exponents of the linear fit for the power-law distribution were higher for periods of high SWRs rate compared to periods of low SWRs rate, indicating a more rapid decay of the power-law distribution, for avalanche size (S) ($p = 6.15 \times 10^{-8}$, Wilcoxon signed-rank test, $n = 30$ epochs from n = 6 animals) and avalanche duration (T) (Wilcoxon signed-rank test, $p = 2.05 \times 10^{-7}$, $n =$ 30 sessions from n = 6 animals). **f,** Deviation from criticality (DCC) calculated as the difference between the linear fit and empirical data of the power-law distribution for periods of low and high SWRs rate. Periods of high SWRs rate had larger DCC compared to periods of low SWRs rate ($p = 0.011$, Wilcoxon signed-rank test (n = 30 sessions from n = 6 animals). Periods of equal total duration were considered for panels **d** to **f**.

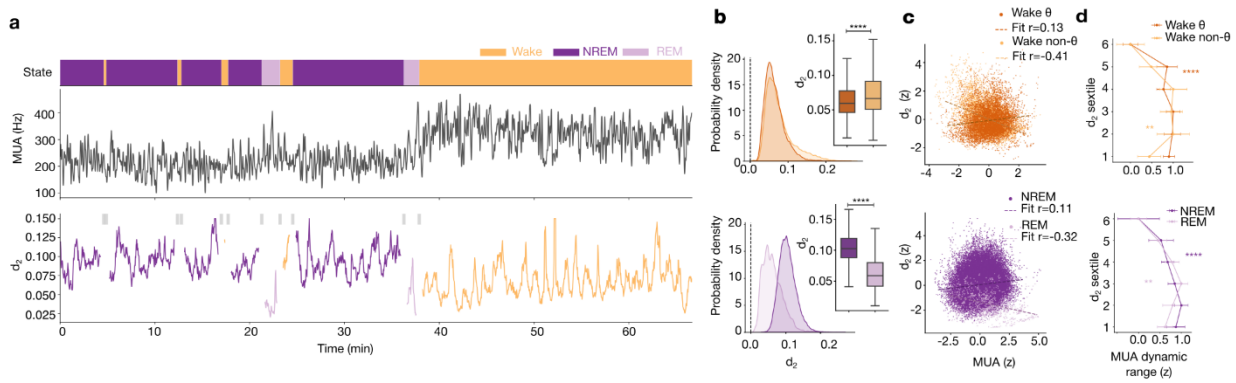


Extended Data Fig.3 a, Mean d_2 was higher during post-task compared to pre-task NREM sleep considering periods of equal SWRs rate (p=0.024, paired Linear Mixed Model, n = 9 from n = 6 animals). **b,** d_2 during post-task and pre-task NREM sleep periods after removing spiking within SWRs ($p < 10^{-15}$, paired Linear Mixed Model, n=9 sessions from n = 6 animals). **c,** d_2 distance correlated with the rate of SWRs even when removing spiking within SWRs (Pearson's: r = 0.85, $p < 10^{-15}$; n = 9 sessions from n = 6 animals). **d,** `Inter-spike-interval (ISI) entropy decreased with SWRs rate (Pearson's: r = −0.80, p = 2.21 × 10`$^{-13}$, n = 9 sessions from n = 6 animals). **e,** d_2 distance displayed a negative correlation with the rate of BARRs even when removing spiking within BARRs (Pearson's: r = -0.86, p =0.028, n = 12 sessions from n = 5 animals). **f,** CA3 pyramidal cell firing was positively correlated with SWRs rate sextiles (Pearson's: r=0.99, $p= 3.5\times10^{-4}$ n = 9 sessions from n = 3 animals). **g,** CA2 pyramidal cell firing was positively correlated with BARRs rate sextiles (Pearson's: r=0.99, $p=5.1\times 10^{-5}$, n = 9 sessions from n = 3 animals).

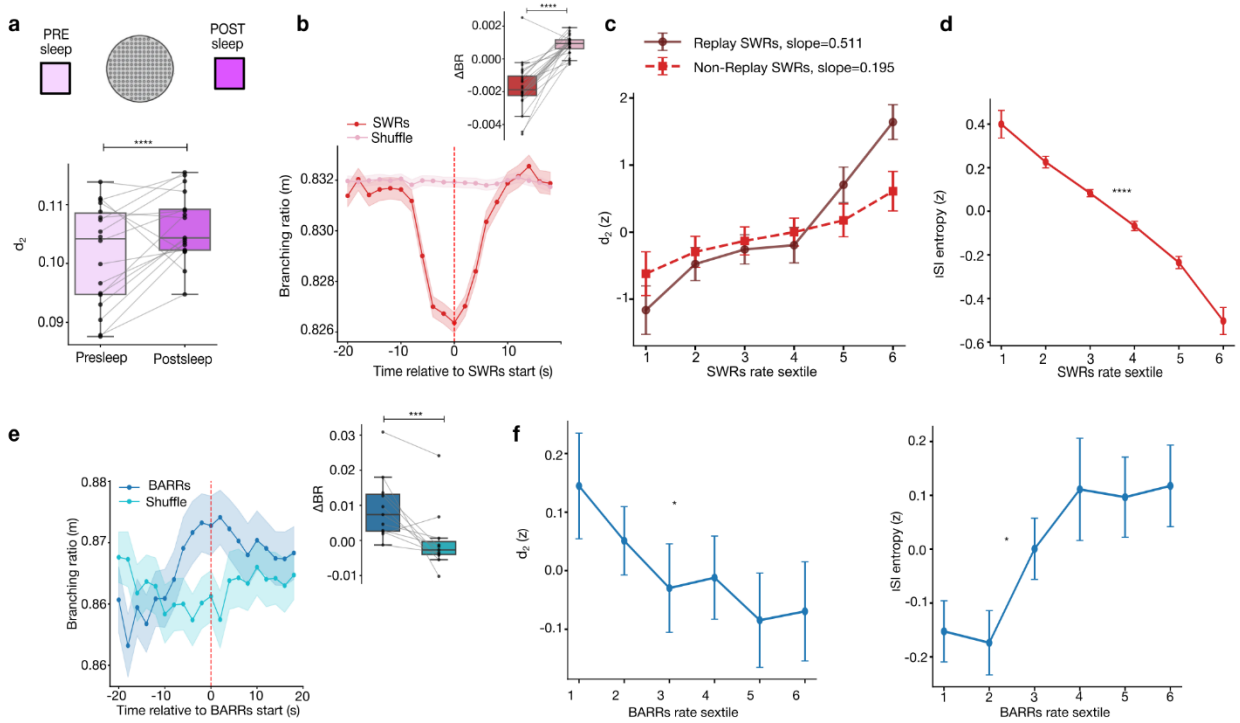

a
CA1
CA2
CA3
SWR
BARR
1mV
50 ms
CA1pyr
CA2pyr
CA3pyr
Putative CCK+
b
E
PV
CCK
CA3
CA1
CA2
CA2 input
CA3 input
MUA Firing rate
c
d
CCK firing rate (z)
PV firing rate (z)
PC firing rate (z)
CA3 input
e
f
g
h
i
j
k
p = 0.7611, F2,87 = 0.27
est.winlen (s)

Extended Data Fig.4. a, Left: example recording from the hippocampus, featuring CA1, CA2 and CA3. Approximate targeting of each corresponding region is indicated by the 300 ms raw traces overlaid on top. Right: example traces of a hippocampal sharp-wave ripple (SWR) and barrage (BARR) recorded in the same session. Event boundaries are indicated by the dashed vertical lines. Traces indicate local field potentials (LFP) and raster plot indicates single units from CA1, CA2, and CA3 pyramidal cells. The firing of putative CCK+ interneurons is also included. **b,** zscored d_2 in relation to input strength colored by MUA firing rate. **c,** CCK firing rate was modulated by both CA3 input (color coded) as well as CA2 inputs (F_{CA2}). Note the negative correlation between CCK firing rate and pyramidal neurons firing for large CA2 input [65] **d,** PV firing rate was positively correlated with pyramidal neurons firing. **(e-g).** Same as in **b-d** but with a larger CCK connectivity. Note that this structure was able to effectively inhibit the network and led to larger control of both CA1 pyramidal neurons and PV firing rates. **(h-j).** Same as in **b-d** but without EE connectivity within CA1. Note that, compared to **b-d**, CA2 input showed a larger control of CA1 pyramidal neurons and PV firing rates. **k,** d_2 estimation was stable for long windows (5, 10 and 15 s) for a stationary process ($p=0.76$, $F_{2,97}=0.27$, ANOVA).

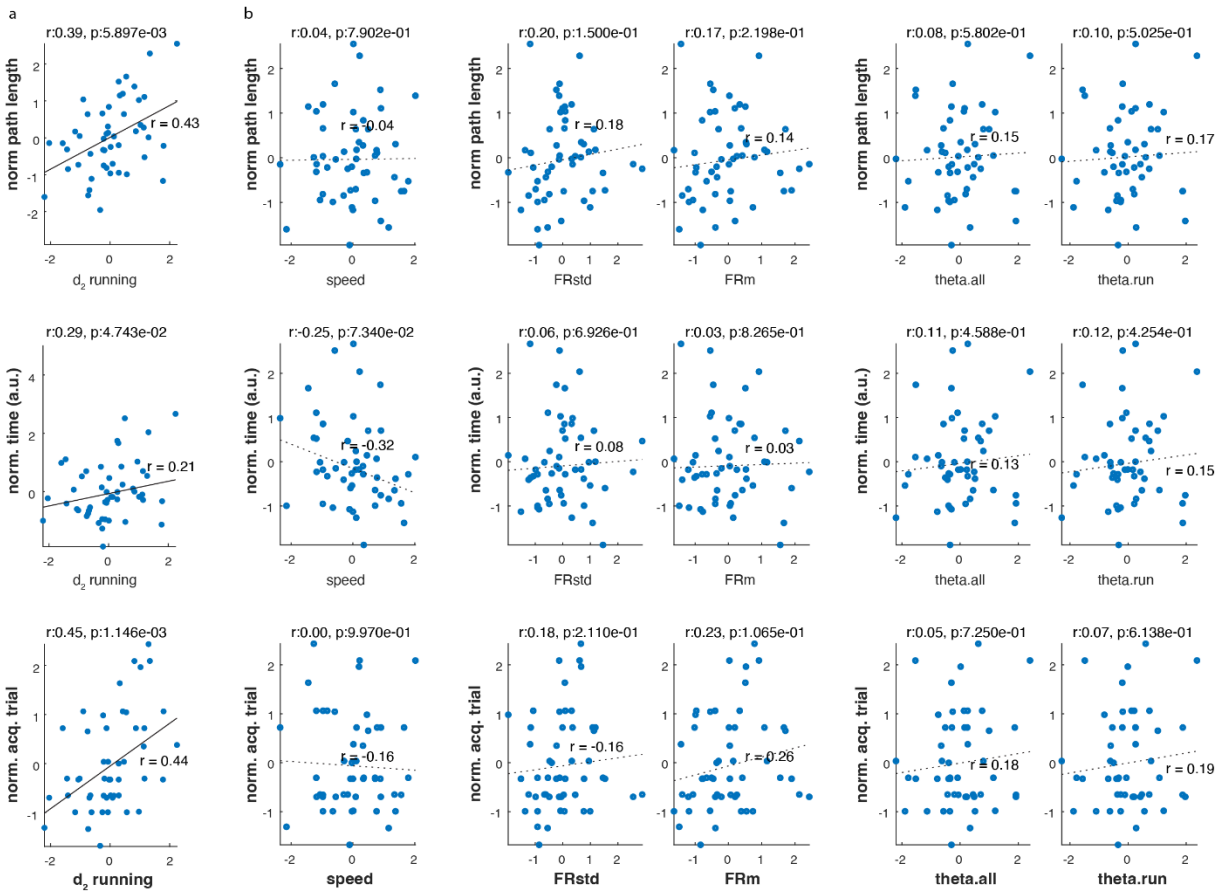
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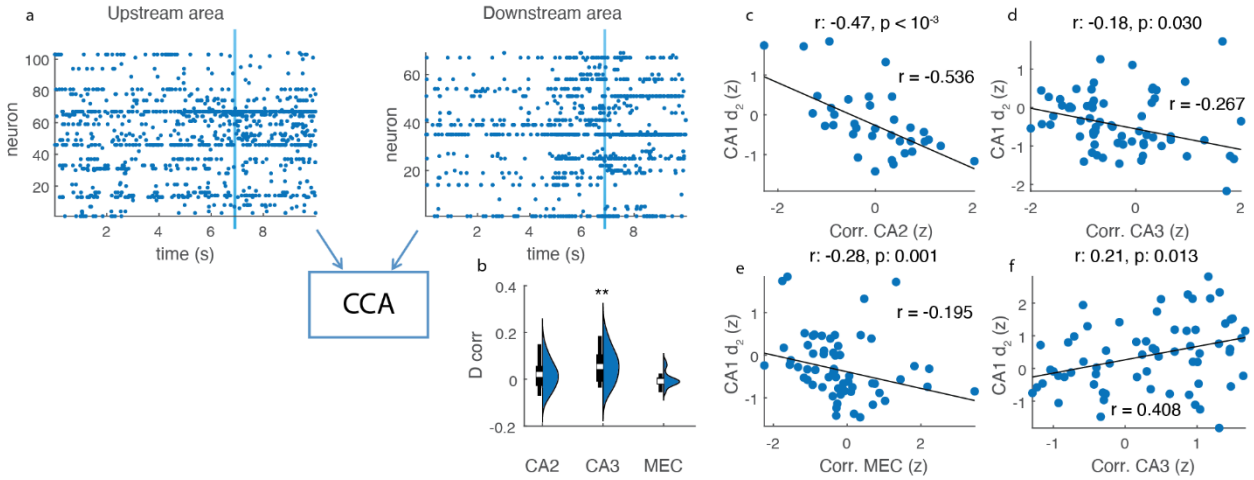
Extended Data Fig.5 **a,** Example of continuous recording illustrating state segmentation (Wake, NREM, REM; top), multiunit firing rate (MUA, middle), and d_2 (bottom) in a rat dataset. **b,** Top: distribution of d_2 during waking for active (theta, θ) and non-active (non-theta, non-θ) periods. Inset boxplots show group comparisons of d_2 value ($p < 1\times10^{-15}$, paired LMM, $n = 20$ sessions from n = 20 sessions in n = 4 animals). Bottom: same for sleeping periods of NREM and REM states. Inset boxplots show group comparisons ($p < 1\times10^{-15}$, LMM, $n = 20$ sessions from n = 4 animals). **c,** Relationship between d_2 and multiunit activity (MUA) during theta and non-theta states during wake (top panel) and REM and NREM stages during sleep (bottom panel). The d_2 relationship with MUA displayed a positive (*r=0.13*) and a negative (r=-0.41) correlation during theta and non-theta wake states respectively. The d_2 relationship with MUA displayed a positive (r=0.11) and a negative (r=-0.32) correlation during NREM and REM sleep stages respectively. Pearson correlations were used for all comparisons (all $p<10^{-15}$), n = 18 sessions from n= 6 animals). **d,** Relationship between MUA dynamic range and d2 sextiles for all states. Note that the dynamic-range increases for lower d_2 for all states (Spearman's: NREM $\rho = -0.43$, $p = 7.51\times10^{-5}$; REM: $\rho = -0.52$, $p = 1.11\times10^{-3}$, Wake θ: $\rho = -0.77$, $p = 3.94\times10^{-9}$. Wake non-θ: $\rho = -0.35$, $p = 9.91\times10^{-3}$). n = 24 sessions from n = 5 animals (NREM), n = 10 sessions from n = 4 animals (REM), n = 13 sessions from n = 5 animals (wake θ) and n= 13 sessions from n = 4 animals (wake non- θ). Note that periods lasting <150 s were excluded from the analysis.



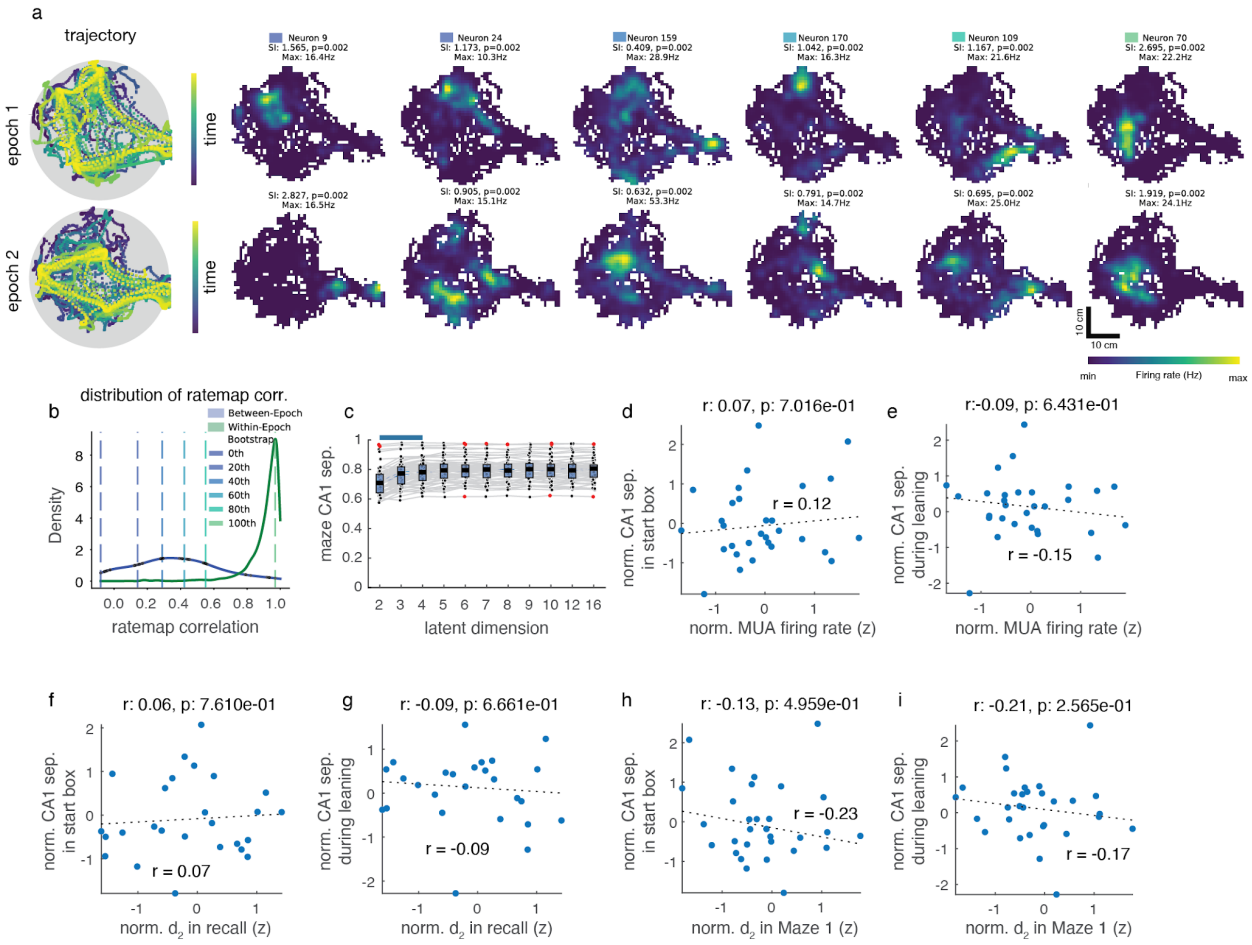
Extended Data Fig.6 a, d_2 during post-task sleep was higher compared to pre-task sleep ($p=5.41 \times 10^{-13}$, paired Linear Mixed Model, n=17 sessions from n = 4 animals). A typical session structure is described in Fig. 4a. **b,** Branching ratio (BR) during SWRs and shuffled periods for an example session. SWRs windows showed a significant decrease in BR compared to shuffles ($p < 1\times10^{-15}$, paired linear mixed model, n=24 sessions from n = 4 animals, paired LMM). **c,** The linear relationship between d_2 and the rate of SWRs associated with replay had a steeper slope than the relationship between d_2 and the rate of SWRs not associated with memory replay (Weighted interaction model: Δslope = 0.316 ± 0.080, p=0.0042, n = 15 sessions in n = 4 animals). **d,** Inter-spike-interval (ISI) entropy decreased with ripple rate (Pearson's: r=0.99, $p =8.88\times10^{-5}$, n=25 sessions from n = 4 animals). **e,** Branching ratio (BR) during BARRs and shuffled periods for an example session. BARRs windows showed a significant increase in BR compared to shuffles ($p= 1.27\times10^{-5}$, paired LMM, n=11 sessions from n = 4 animals). **f,** d_2 decreased with barrage rate (Pearson's: r= -0.91, p = 0.011, n=12 sessions from n = 4 animals). **g,** Inter-spike-interval (ISI) entropy increased with BARRs rate (Pearson's: r= 0.91, p = 0.012 (n=12 session from n = 4 animals).



Extended Data Fig.7 a, Correlation between learning performance and distance to criticality during the early phase (first 5 trials) of the learning task. Learning performance is measured as the average path length (1st row), average time spent to collect all rewards (2nd row), and number of trials to collect all rewards (3rd row) during the early phase (first 5 trials) (n = 49 sessions from 3 animals). Significant and non-significant correlations are fitted with solid and dashed lines respectively. Spearman values are indicated above the lines. **b,** Correlations between learning performance (measured as indicated in **a** and average speed, mean multiunit firing rate, and standard deviation of multiunit firing rate during maze time are shown in first, second and third columns respectively. Correlations between learning performance (measured as indicated in **a**) and LFP theta power during maze time (fourth column) and during running periods (fifth column). Significant and non-significant correlations are fitted with solid and dashed lines respectively. Spearman values are indicated above the lines. Data is normalized within each animal.

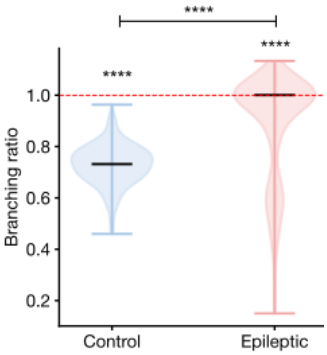


Extended Data Fig.8 a, Schematic of cross-regional coordination measured by population correlation using the Canonical Correlation Analysis (CCA) method. **b,** CA3-CA1 population correlation significantly increased during learning (CA2: p = 0.15, CA3: p = 0.005, MEC: p = 0.38, Wilcoxon signed-rank test). **c,** Correlation between CA1 d_2 during the earlier (first 5 minutes) part of the task and population correlation between CA1 and CA2 ($p<0.001$, Spearman correlation, n = 12 sessions from n = 3 animals). **d,** Same as in **c** but for CA3 correlation ($p<0.05$, Spearman correlation, n = 13 sessions from n = 3 animals). **e,** Same as in **c** but for MEC correlation ($p<0.01$, Spearman correlation, n = 12 sessions from n = 2 animals). **f,** Same as in **d** Correlation between CA1 d_2 during the later (last 5 minutes) part of the task and population correlation between CA1 and CA3 ($p<0.05$, Spearman correlation, n = 13 sessions from n = 3 animals). Significant and non-significant correlations are fitted with solid and dashed lines respectively. Spearman values are indicated above the lines. Data is normalized within each animal.



Extended Data Fig.9 a, Left, example of animal trajectories during the first configuration (top) and second configuration (bottom) of the task. Right, firing maps of representative hippocampal neurons during the first configuration (top) and second configuration (bottom) of the task. **b,** Quantification of the density of rate map changes for the hippocampal population across maze configurations (mean ± SD across configurations: 0.3619±0.2507; a within configuration 100 times bootstrap is generated as a control for each neuron: 0.9109±0.1045, Mann-Whitney U test: $p < 10^{-15}$). **c,** Separability of latent task manifold is plotted for increasing number of latent dimensions. Note that a dimension higher than 5 is not significantly different from 16 latent dimensions ($p < 10^{-15}$, Friedman's test followed by Dunn & Sidák's approach). **d,** Correlation between CA1 latent embedding for the two configurations and mean multinunit firing rate during the start-box periods of the learning phase of the task (n=30 sessions from n = 3 animals). **e,** Correlation between CA1 latent embedding separability for the two configurations and mean multinunit firing rate during the trials in the learning phase of the task (n=30 sessions from n = 3 animals). **f,** Correlation between CA1 latent embedding separability for the two configurations and distance to criticality during start-box periods of the recall test (of the previous maze) (n=30 sessions from n = 3 animals). **g,** Correlation between CA1 latent embedding separability for the two configurations and distance to criticality during trial periods of the recall test (of the previous maze) (n=30 sessions from n = 3 animals). Significant and non-significant correlations

are fitted with solid and dashed lines respectively. Spearman values are indicated above the lines. Data is normalized within each animal.



Extended Data Fig.10 Branching ratio (m) in control sessions (blue) and epileptic sessions (red). Note a higher branching ratio for epileptic compared to control sessions ($p < 10^{-15}$, Mann–Whitney test) and a branching ratio in the supercritical regime (m>1) for epileptic sessions ($p < 10^{-15}$).